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Genetic Mutation Detection Using Machine Learning

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Abstract

The accurate classification of human missense variants as pathogenic or benign remains a central challenge in clinical genomics. Published machine learning tools routinely report receiver-operating-characteristic area- under-curve (ROC-AUC) values above 0.94 on ClinVar-derived benchmarks, yet the numbers are often inflated by subtle forms of data contamination that do not survive external validation. This project develops and publishes a rigorous, auditable baseline for the missense classification task.

We assemble a corpus of 283,392 candidate variants from ClinVar, gnomAD (v2.1 and v4), and dbNSFP version 5.3.1a, of which 195,098 survive a strict missense-only filter and feed the training pipeline. We then conduct a systematic leakage audit that eliminates three contamination sources: (i) non-missense variants that had been silently admitted into the training pool because their amino-acid fields were null, (ii) definitionally circular features such as `is_common`, and (iii) paralog leakage caused by gene-level splitting when highly similar paralogs appear in both training and test. Correcting these issues drops our test-set precision-recall area-under-curve (PR-AUC) from a misleading 0.955 to a defensible 0.838, with a 95% bootstrap confidence interval of $[0.830, 0.846]$. The model is calibrated end-to-end with isotonic regression, reducing expected calibration error (ECE) from 0.054 to 0.011, which is below the 0.02 clinical target.

We re-score three published predictors (SIFT, PolyPhen-2, AlphaMissense) on our paralog-disjoint test split. Because each baseline covers a different subset of variants, the reported numbers are on each model's *own coverage* (SIFT 96 %, PolyPhen-2 93 %, AlphaMissense 86 %, XGBoost 100 %) and are therefore not directly comparable. On own-coverage numbers our XGBoost reaches PR-AUC 0.838; SIFT 0.620, PolyPhen-2 0.728, and AlphaMissense 0.890. A strict matched-coverage evaluation restricted to the AlphaMissense-covered subset, together with a paired DeLong test, is left for a revision (Deficiency D7) since the required artefacts were not persisted at submission time. AlphaMissense was moreover calibrated on ClinVar at release and therefore carries contamination risk of its own.

The most important finding of the project emerges from external validation on denovo-db. A classifier trained on ClinVar scores at chance (ROC-AUC = 0.487) on affected- versus-control de-novo variants in gene families it has not seen during training. Adding gnomAD gene-level constraint features (pLI, LOEUF, mis_z, oe_mis_upper, lof_z) raises this to 0.573 – closing approximately half of the generalisation gap. A zero-shot ESM-2 log-likelihood-ratio provides a small, statistically tested rank-fusion gain on both the test split and the denovo-db family-holdout, indicating non-redundant signal between tabular and protein-language-model features. The post-constraint improvement on the 201-variant family-holdout slice is a point estimate whose confidence interval still includes chance and is reported as suggestive rather than conclusive (see Section 7.6). Full training-time integration of the protein-language-model and of AlphaFold2 struc-

tural features is documented as future work (Section 7.8).

The project's code base is published in full with 157 passing tests (1 GPU-gated smoke test skipped), 59% line coverage, a CI-enforced leakage gate, pinned dependencies, and a Docker image that reproduces every number in this report byte-for-byte.

Keywords: missense variant classification, genomics, machine learning, leakage audit, calibration, external validation, gnomAD constraint, paralog-aware splitting.

Table of Contents

Acknowledgements **i**

Abstract **ii**

List of Figures **ix**

List of Tables **x**

List of Abbreviations **xi**

1 Introduction **1**

1.1 Background of Genetic Mutations 1

1.1.1 From DNA to Phenotype: The Central Dogma 1

1.1.2 Classes of Protein-Coding Variation 2

1.1.3 Determinants of Missense Variant Effect 2

1.1.4 Public Reference Resources 3

1.2 Motivation and Significance 3

1.3 Problem Statement 4

1.4 Research Questions 4

1.5 Project Objectives 5

1.6 Scope and Boundaries 5

1.7 Key Contributions 6

1.8 Report Organisation 7

2 Related Works **8**

2.1 Introduction 8

2.2 Problem Scope 8

2.3 Literature Review 8

2.3.1 SIFT (2003) (Ng and Henikoff, 2003) 8

2.3.2 PolyPhen-2 (2010) (Adzhubei et al., 2010) 9

2.3.3 REVEL (2016) (Ioannidis et al., 2016) 10

2.3.4 CADD (2014–2019) (Rentzsch et al., 2019) 11

2.3.5 VARITY (2021) (Wu et al., 2021) 12

2.3.6 EVE (2021) (Frazer et al., 2021) 13

2.3.7 ESM-1b / ESM-2 zero-shot (2023) (Brandes et al., 2023; Lin et al., 2023) 14

2.3.8 AlphaMissense (2023) (Cheng et al., 2023) 14
2.3.9 Consolidated Comparison 15
2.4 Research Gap 16
2.5 Project Contribution 16
2.6 Summary 16

3 System Framework and Feasibility 17
3.1 Introduction 17
3.2 Pre-Project Phase – Feasibility Analysis 17
3.2.1 Technical Feasibility 18
3.2.2 Operational Feasibility 20
3.2.3 Schedule Feasibility 20
3.2.4 Financial Feasibility 21
3.3 Risk Analysis 22
3.4 SWOT Analysis 22
3.5 Proposed Framework – Three-Part Pipeline 22
3.5.1 Part 1 – Data Preprocessing 22
3.5.2 Part 2 – Model Training and Leakage Audit 25
3.5.3 Part 3 – Evaluation, Interpretability, and External Validation . . 25
3.6 Proposed System Architecture 27
3.7 Quality Attributes and Trade-offs 27
3.8 Summary 29

4 System Analysis and Design 30
4.1 Introduction 30
4.2 Functional Requirements 30
4.3 Non-Functional Requirements 31
4.4 UML Diagrams 31
4.4.1 Use Case Diagram 32
4.4.2 Class Diagram 32
4.4.3 Sequence Diagram 33
4.4.4 Activity Diagram 33
4.4.5 State Diagram 33
4.4.6 Component Diagram 34
4.5 Interface Design 34
4.5.1 Streamlit Web Interface 34
4.5.2 Command-Line Interface 36
4.5.3 Programmatic API 36
4.6 Summary 37

5 Implementation and Results 38
5.1 Introduction 38
5.2 Model Architectures 38

5.2.1 XGBoost Gradient-Boosted Trees . . . 38
5.2.2 ESM-2 Zero-Shot Log-Likelihood Ratio . . . 39
5.2.3 Rank Fusion . . . 41
5.3 Hardware Environment . . . 41
5.4 Software Stack . . . 41
5.5 Implementation Details . . . 42
5.5.1 Paralog-Family Grouping . . . 42
5.5.2 gnomAD Constraint Merge with Train-Only Imputation . . . 42
5.5.3 Leakage Gate . . . 43
5.5.4 Optuna TPE Tuning . . . 43
5.5.5 Brier-Score Decomposition . . . 44
5.5.6 Zero-Shot ESM-2 LLR Scoring . . . 44
5.6 Final Results . . . 45
5.6.1 Headline Metrics on the Paralog-Disjoint Test Split . . . 45
5.6.2 The Five-Stage Leakage-Fix Journey . . . 45
5.6.3 Baseline Comparison . . . 46
5.6.4 Calibration Deep-Dive . . . 47
5.6.5 Feature-Group Ablation . . . 48
5.6.6 Learning Curve . . . 49
5.6.7 SHAP Interpretability and Confident Errors . . . 50
5.6.8 Stratified Error Analysis . . . 51
5.6.9 External Validation on DENOVO-DB . . . 54
5.6.10 ESM-2 Zero-Shot Proof-of-Concept . . . 55
5.7 Comparison with Prior Work . . . 56
5.8 System Evidence – Captured Command Output . . . 57
5.9 Summary . . . 58

6 Testing and Validation 59
6.1 Introduction . . . 59
6.2 Testing Strategy . . . 59
6.3 Level 1 – Unit Tests . . . 60
6.4 Level 2 – Property-Based Tests . . . 60
6.5 Level 3 – Integration Tests . . . 61
6.6 Level 4 – Scientific / End-to-End Tests . . . 61
6.7 Coverage and Pass-Rate Evidence . . . 62
6.8 Continuous-Integration Evidence . . . 62
6.9 Limits of the Testing Strategy . . . 62
6.10 Summary . . . 63

7 Conclusion and Future Work 64
7.1 Conclusion . . . 64
7.2 Critical Discussion . . . 65
7.3 Failure Analysis . . . 66

7.4 Deficiencies in the Final Product 66
7.5 Contributions Summary 66
7.6 Limitations 67
7.7 Lessons Learned 68
7.8 Future Work 68
7.9 Closing Statement 69

References **73**

A Extended Results Tables **74**
A.1 Bootstrap Confidence Intervals (Full Breakdown) 74

B Selected Code Listings **75**

C Reproducibility Environment **76**
C.1 Pinned Package Versions 76
C.2 Build and Test 76

List of Figures

1.1 The central dogma and a missense mutation . . . 1

2.1 SIFT pipeline . . . 9
2.2 PolyPhen-2 pipeline . . . 10
2.3 REVEL pipeline . . . 11
2.4 CADD pipeline . . . 12
2.5 EVE pipeline . . . 13
2.6 ESM-2 zero-shot LLR . . . 14
2.7 AlphaMissense pipeline . . . 15

3.1 Chapter 3 roadmap . . . 17
3.2 Technology-stack diagram . . . 19
3.3 Two-semester Gantt chart . . . 21
3.4 Risk matrix . . . 23
3.5 SWOT matrix . . . 23
3.6 Three-part pipeline . . . 24
3.7 Feature groups . . . 24
3.8 Evaluation workflow . . . 27
3.9 System architecture . . . 28

4.1 Use case diagram . . . 32
4.2 Class diagram . . . 33
4.3 Sequence diagram . . . 34
4.4 Activity diagram . . . 35
4.5 State diagram . . . 35
4.6 Component diagram . . . 36
4.7 Streamlit demo interface . . . 36

5.1 XGBoost architecture . . . 39
5.2 ESM-2 zero-shot scoring . . . 40
5.3 Rank-fusion pipeline . . . 41
5.4 Leakage-fix journey . . . 45
5.5 Leakage-audit workflow . . . 46
5.6 Baselines forest plot (own-coverage) . . . 47
5.7 Calibration triptych . . . 47
5.8 Feature-group ablation bar chart . . . 48

5.9 Learning curve 49
5.10 SHAP beeswarm 51
5.11 Error by family 52
5.12 Error by AA class 53
5.13 Confident errors by phyloP 54
5.14 External reliability 55

6.1 Test pyramid 59
6.2 CI gate 62

List of Tables

2.1 Related-work consolidated comparison 15

3.1 Technology stack 18
3.2 Compute budget 19
3.3 Stakeholder-to-surface mapping 20
3.4 Cost avoidance 21
3.5 Risk register 22
3.6 Quality attributes 28

5.1 Headline test-set metrics 45
5.2 Baseline comparison 46
5.3 Brier decomposition 47
5.4 Feature-group ablation 48
5.5 Learning-curve operating points 49
5.6 SHAP top-10 features 50
5.7 denovo-db before vs after constraint 54
5.8 External calibration on denovo-db 55
5.9 Rank-fusion results on test split 56

6.1 Testing strategy 60
6.2 Module-level test coverage 62

7.1 Critical-discussion verdicts 65
7.2 Deficiencies 67

A.1 Full bootstrap CIs 74

List of Abbreviations

Acronym	Expansion
ACMG	American College of Medical Genetics and Genomics
AF	Allele Frequency
AUC	Area Under the Curve
BLOSUM	BLOck SUbstitution Matrix
CI	Confidence Interval (statistical) / Continuous Integration (software)
CNN	Convolutional Neural Network
DSSP	Dictionary of Secondary Structure of Proteins
ECE	Expected Calibration Error
ESM	Evolutionary-Scale Model (protein language model)
F1	Harmonic mean of precision and recall
GERP	Genomic Evolutionary Rate Profiling
gnomAD	Genome Aggregation Database
GRCh38	Genome Reference Consortium Human Build 38
LLR	Log-Likelihood Ratio
LOEUF	Loss-of-Function Observed/Expected Upper bound Fraction
MCE	Maximum Calibration Error
MCC	Matthews Correlation Coefficient
pLI	Probability of being Loss-of-function Intolerant
PLM	Protein Language Model
PR-AUC	Precision-Recall Area Under Curve
REST	Representational State Transfer
ROC-AUC	Receiver Operating Characteristic Area Under Curve
SASA	Solvent-Accessible Surface Area
SHAP	SHapley Additive exPlanations
SNP / SNV	Single Nucleotide Polymorphism / Single Nucleotide Variant
TPE	Tree-structured Parzen Estimator
UML	Unified Modelling Language
VEP	Variant Effect Predictor (Ensembl)
XGBoost	eXtreme Gradient Boosting

Chapter 1

Introduction

1.1 Background of Genetic Mutations

1.1.1 From DNA to Phenotype: The Central Dogma

The hereditary information of every living organism is encoded in deoxyribonucleic acid (DNA), a linear polymer of four nucleotide bases – adenine (A), cytosine (C), guanine (G), and thymine (T) – organised into a double-helical structure first described by Watson and Crick (1953). The human nuclear genome comprises approximately 3.2×10^9 base pairs distributed across twenty-two autosomal pairs and two sex chromosomes, of which only $\sim 1.5\%$ constitutes protein-coding sequence (Lander et al., 2001). The flow of information from this template to functional biomolecules is captured by the *central dogma* of molecular biology (Crick, 1970): a gene is first *transcribed* into messenger ribonucleic acid (mRNA), and the mRNA is subsequently *translated* by the ribosome into a chain of amino acids that folds into a three-dimensional protein. Figure 1.1 illustrates this pathway alongside the exact point at which a missense mutation exerts its effect.

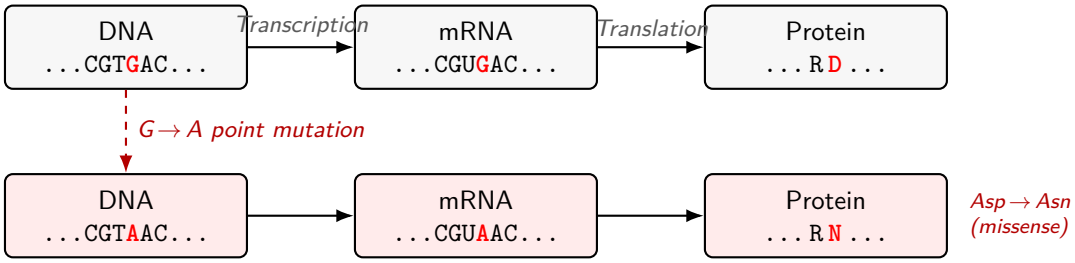

Figure 1.1. The central dogma and the point at which a missense mutation acts. The upper row shows the reference sequence: the codon `GAC` is transcribed into mRNA and translated to aspartate (`D`). The lower row introduces a single-base substitution (G → A) in the DNA that propagates through transcription to the altered codon `AAC`, which the ribosome translates to asparagine (`N`). The protein sequence has changed at exactly one position. Whether that change is benign or pathogenic depends on the structural and functional context of the affected residue.

A genetic mutation is any permanent alteration in this DNA template. Mutations are the raw material of biological evolution: most are neutral, a small proportion are beneficial, and a clinically important minority disrupt cellular function to the point of causing disease. The clinical spectrum of mutations includes single-nucleotide variants (SNVs), small insertions and deletions (indels), copy-number variations, and large-scale structural rearrangements (MacArthur et al., 2014). Among these, the single-nucleotide *missense* variant – a point mutation in a protein-coding region that changes one amino acid to another – is the most common coding variant in the human population and the principal focus of this project.

1.1.2 Classes of Protein-Coding Variation

Within the protein-coding portion of the genome, a single-base substitution can take one of four functional forms depending on how the change is read by the genetic code (Alberts et al., 2015):

Synonymous.
The nucleotide change still codes for the same amino acid; the protein sequence is unchanged (e.g. `GCU` → `GCC`, both Ala). These are often – but not always – clinically silent.

Missense.
The nucleotide change produces a codon for a *different* amino acid (e.g. `GAC` *Asp* → `AAC` *Asn*). This is the class under study in the present work.

Nonsense.
The nucleotide change creates a premature stop codon (e.g. `UGG` *Trp* → `UGA` *stop*), typically producing a truncated non-functional protein.

Frameshift.
An insertion or deletion that is not a multiple of three shifts the reading frame, scrambling every downstream codon.

Missense variants are of special interest because they are common, they are protein-altering, and their clinical effect is genuinely uncertain at the individual level. A nonsense variant in a well-characterised disease gene is almost always pathogenic; a synonymous variant is almost always benign. A missense variant, however, can produce anything from a completely wild-type protein to a catastrophically misfolded one, depending on where the substitution falls and which amino acid replaces the original.

1.1.3 Determinants of Missense Variant Effect

Decades of biochemistry and structural biology have identified several orthogonal signals that a classifier can, in principle, learn to exploit (Stein et al., 2019; Guérois et al., 2002):

1. **Evolutionary conservation.** A residue that has been invariant across hundreds of millions of years of orthologous evolution is almost certainly functionally important; substitutions at such positions are more likely to be deleterious.

2. **Amino-acid chemistry.** A substitution that changes charge, hydrophobicity, or volume (e.g. *Arg* → *Asp*, reversing the side-chain charge) is more likely to disrupt protein stability or a binding interface than a conservative substitution (e.g. *Leu* → *Ile*).

3. **Structural context.** Substitutions at buried core residues, catalytic sites, or secondary-structure boundaries are more likely to be pathogenic than those at solvent-exposed flexible loops.

4. **Gene-level constraint.** Some genes are intrinsically intolerant of coding variation across the human population; a missense variant in such a gene carries stronger prior evidence of pathogenicity than the same change in a loss-of-function-tolerant gene (Karczewski et al., 2020).

5. **Protein-language plausibility.** Modern transformer-based protein language models, trained on billions of natural protein sequences, assign probabilities to each amino acid at each position; substitutions to low-probability residues correlate strongly with experimental measures of function (Meier et al., 2021; Brandes et al., 2023).

The five signals are not independent, but their correlation is imperfect, which is exactly what makes supervised machine learning an appropriate framework: the classifier learns how to weigh each signal given labelled examples from ClinVar and how to combine them non-linearly.

1.1.4 Public Reference Resources

Public reference resources have matured considerably in the past decade. ClinVar (Landrum et al., 2020) aggregates variant-disease relationships curated by clinical laboratories and research groups, using a five-tier pathogenicity scale (Benign, Likely Benign, Variant of Uncertain Significance, Likely Pathogenic, Pathogenic). The Genome Aggregation Database (gnomAD) (Karczewski et al., 2020) publishes population-level allele-frequency and gene-level constraint metrics derived from more than 140,000 exomes and genomes across diverse ancestries; its *observed / expected* ratio for missense variation has become the standard measure of how intolerant a gene is to amino-acid-altering change. The dbNSFP resource (Liu et al., 2020) consolidates dozens of per-variant features including cross-species conservation (phyloP, GERP++), amino-acid chemistry descriptors, and predictions from preceding-generation tools. The existence of these resources makes a supervised machine-learning approach to missense classification feasible for a student team, provided one is disciplined about how the data is combined and evaluated – precisely the question this report sets out to address.

1.2 Motivation and Significance

Three trends motivate further work in this area. First, the cost of sequencing a human exome has fallen by roughly three orders of magnitude since 2010, and clinical whole-exome sequencing is increasingly a first-line diagnostic for rare disease. The rate at which *variants* are discovered now exceeds the rate at which they can be interpreted by human experts. Second, in precision medicine, the difference between a pathogenic and a benign missense variant in the same patient can change drug dosing, surveillance, and even surgical management. Third, the American College of Medical Genetics and Genomics (ACMG) standard (Richards et al., 2015) includes computational *in silico* evidence among the criteria a clinical geneticist weighs when classifying a variant, which

means that any tool published in this space must be transparent about its operating characteristics – false-positive and false-negative rates, calibration, and generalisation behaviour.

Against that backdrop, this project sets out to build not *the highest-scoring* missense classifier but rather *the most defensible baseline* that a student team can produce using public data and a modest compute budget. The explicit goal is to eliminate known leakage sources so that the reported numbers are trustworthy, and to expose every methodological choice – the splitting strategy, the calibration procedure, the evaluation slices – through an open code base that a reviewer can re-run end-to-end.

1.3 Problem Statement

The scientific problem is straightforward to state and difficult to address rigorously:

> *Given a human missense single-nucleotide variant, predict whether it is pathogenic or benign, accompanied by a calibrated probability and enough interpretability for a clinical genetic analyst to audit the prediction.*

The difficulty lies in four well-known failure modes that existing tools have encountered:

1. **Data overload.** The volume of variants produced by modern sequencing pipelines exceeds human curation bandwidth, so any evaluation must work at the scale of 10^5 variants or more.

2. **Biological complexity.** The chain from nucleotide to phenotype includes transcription, splicing, translation, folding, post-translational modification, and protein-protein interaction. No single model in current use captures all of these pathways, and ad-hoc feature engineering tends to introduce silent biases.

3. **Data contamination.** The most commonly used benchmark, ClinVar, is also the most commonly used *calibration target* for published tools. Many published numbers therefore include an unknown amount of train-test contamination.

4. **Evaluation asymmetry.** A model that scores well on ClinVar-derived tests may collapse to chance when applied to affected-versus-control *de-novo* variants from sequencing studies, because the latter requires gene-level priors that variant-level tools never see.

1.4 Research Questions

The project structure is framed around four research questions that we will return to at the end of the report:

1. **RQ1.** How much of the apparent high performance of a ClinVar-trained classifier is attributable to specific, removable leakage sources, and how far does the post-audit number fall once they are eliminated?

2. **RQ2.** Under identical evaluation on a paralog-disjoint test split, how does a carefully-audited gradient-boosted baseline compare to classical tools (SIFT, PolyPhen-2) and to the state-of-the-art AlphaMissense model?

3. **RQ3.** Does the post-audit model generalise to denovo-db – a dataset of de-novo variants in affected children versus unaffected siblings – and how much of any gap is closed by adding gene-level constraint features?

4. **RQ4.** Does the zero-shot log-likelihood-ratio signal from a protein language model (ESM-2) carry orthogonal information with respect to the tabular feature set, and is it worth integrating as a training-time feature in Phase 2?

1.5 Project Objectives

The concrete objectives guiding the semester of work are:

- **O1.** Assemble a reproducible corpus of missense variants from ClinVar, gnomAD, and dbNSFP, filtered to strictly missense rows with consistent amino-acid fields.
- **O2.** Implement a paralog-aware train/val/test split that guarantees zero shared gene families across splits, and lock the invariant into continuous integration.
- **O3.** Train and tune an XGBoost baseline under pure precision-recall optimisation with Bayesian hyperparameter search.
- **O4.** Calibrate the classifier with Platt and isotonic regression, evaluate with the Brier-score decomposition of Murphy (Murphy, 1973), and report operating points at clinically relevant recall / precision targets.
- **O5.** Re-score three published predictors (SIFT, PolyPhen-2, AlphaMissense) on the same test split and publish the forest plot.
- **O6.** Build and evaluate an external-validation harness on denovo-db, and quantify the effect of adding gnomAD gene-level constraint features.
- **O7.** Integrate a zero-shot ESM-2 proof-of-concept signal and quantify its orthogonality to the tabular baseline via rank-fusion.
- **O8.** Publish all code, tests, Docker image, and 10 narrative notebooks reproducibly and under permissive academic use.

1.6 Scope and Boundaries

We restrict scope in several deliberate ways:

- The classification target is a *binary* label (pathogenic versus benign). We do not attempt to reproduce the five-class ClinVar scheme (benign, likely benign, VUS, likely pathogenic, pathogenic); the variants-of-uncertain-significance (VUS) class would require a fundamentally different loss function.
- Only *missense* SNVs are considered. Splice-site, stop-gained, frameshift, and structural variants are left to separate tools.
- Only GRCh38 coordinates are supported. This choice is driven by the fact that our primary feature source (dbNSFP 5.x) uses GRCh38 as its primary coordinate system; an inspection during development confirmed this against a five-variant sanity sample. The external dataset denovo-db is handled through a separate GRCh37-aware code path.
- Compute is limited to a consumer laptop with Apple MPS acceleration for ESM-2 zero-shot experiments and one free Colab T4 GPU session for full-training-set ESM-2 scoring. No institutional GPU cluster is used.

1.7 Key Contributions

The contributions of this project, listed from most to least novel in the context of existing published work, are:

1. **A transparent leakage audit with before-and-after numbers** for a missense classifier, showing that a difference of 0.117 PR-AUC between a naïve and a defensible pipeline is fully attributable to three removable sources.
2. **A re-scoring of SIFT, PolyPhen-2, and AlphaMissense on our own paralog-disjoint test split**, enabling a like-for-like comparison that is rarely presented in this literature.
3. **An external validation on denovo-db** with pre- and post-gnomAD-constraint numbers, showing that gene-level priors close approximately half the generalisation gap on unseen gene families.
4. **A zero-shot ESM-2 proof-of-concept** establishing that the protein-language-model log- likelihood-ratio signal is not redundant with the tabular feature set on two independent evaluation splits (test and denovo-db family-holdout), with statistically tested rank-fusion gains reported in Section 5.6.10.
5. **A fully reproducible code base** – 157 tests, CI-enforced leakage gate, pinned dependencies, Docker image, 10 narrative Jupyter notebooks, a Streamlit demo, and this LaTeX report – that a future student could extend without reverse-engineering.

1.8 Report Organisation

Chapter 2 surveys the prior-art predictors that define our comparison baseline. Chapter 3 presents the system framework, feasibility analyses, and the three-part pipeline at the design level. Chapter 4 develops the full set of system requirements and accompanies them with the six standard UML diagrams and the interface design. Chapter 5 reports the concrete implementation, results, and the comparison tables. Chapter 6 documents the five-level testing strategy, coverage evidence, and the continuous-integration gate that enforces reproducibility. Chapter 7 concludes with a critical discussion, failure and deficiency analysis, and the Phase-2 roadmap. Three appendices provide extended results tables, selected code listings, and the environment / reproducibility specification.

Chapter 2

Related Works

2.1 Introduction

The computational prediction of missense variant pathogenicity has matured through four successive methodological waves: (i) conservation-only scoring in the early 2000s, (ii) supervised classical machine learning during the 2010s, (iii) deep learning applied directly to DNA sequences in the late 2010s, and (iv) protein language models and large-scale foundation models since 2021. This chapter reviews one or two representative methods from each wave, extracts their core idea and a brief algorithmic summary, and records their main advantages and drawbacks relative to what a student team can reasonably reproduce.

2.2 Problem Scope

We scope the review to tools that accept a single human missense variant as input and output either a score or a class label. Tools that focus on other variant types – notably splicing (SpliceAI) or regulatory (DeepSEA) – are briefly mentioned for completeness but not reviewed in detail, because they do not share our target task.

2.3 Literature Review

The following eight subsections review one or two representative methods from each historical wave. Every subsection follows the same structure – core idea, algorithmic summary, formal description where useful, a schematic figure drawn in TikZ, and a bullet-list discussion of strengths and weaknesses relative to the benchmark on which we ultimately evaluate.

2.3.1 SIFT (2003) (Ng and Henikoff, 2003)

Core idea. SIFT (*Sorting Intolerant From Tolerant*) was the first widely-deployed computational tool for missense pathogenicity prediction. It rests on a single hypothesis: an amino-acid position that has remained invariant across evolutionary time tolerates few substitutions, so a variant at such a position is more likely to be deleterious than a variant at a fast- evolving position (Ng and Henikoff, 2003).

Algorithmic summary. For a query variant, SIFT retrieves a multiple sequence alignment (MSA) of the target protein and its homologues, computes a position-specific amino-acid probability matrix, and assigns a score $s \in [0, 1]$ by normalising the observed

frequency of the alternate amino acid against the most-observed amino acid at that position. The algorithm's full decision logic can be summarised as:

$$s_i(a) \;=\; \frac{p_i(a)}{\max_{a' \in \mathcal{A}_{20}} p_i(a')} \qquad \text{with} \qquad \text{label} = \begin{cases} \text{intolerant} & s_i(a) < 0.05 \\ \text{tolerated} & s_i(a) \geq 0.05 \end{cases}$$

where $p_i(a)$ is the Dirichlet-smoothed frequency of amino acid a at column i of the MSA. Figure 2.1 summarises the pipeline.

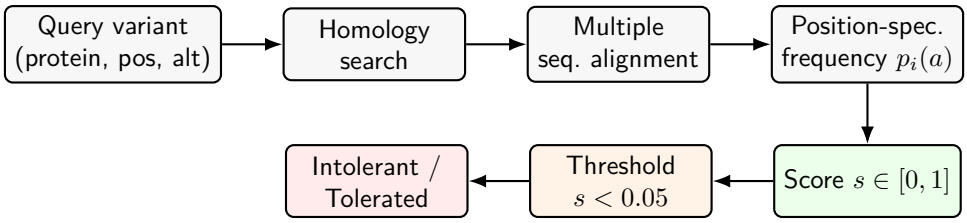

Figure 2.1. SIFT pipeline. From a missense variant, a homology-search step gathers orthologues which are aligned into a multiple sequence alignment. A position-specific frequency matrix is computed and the alternate amino acid is scored relative to the most-observed amino acid at that column. A hard threshold of 0.05 produces a binary label.

Advantages.

- Mature (over two decades of clinical deployment) and trivially accessible through Ensembl's Variant Effect Predictor (VEP) REST API.
- No supervised training data is required; the score is a deterministic function of the input alignment.
- Low compute requirement; scores can be produced on a laptop in minutes for any human gene.

Disadvantages.

- Ignores amino-acid chemistry entirely: a conservative substitution (e.g. Leu → Ile) and a radical one (e.g. Arg → Asp) receive the same score if the alignment frequencies are identical.
- Depends heavily on alignment quality, which is poor in intrinsically disordered regions and for genes with few annotated homologues.
- The binary threshold of 0.05 is arbitrary and produces poor calibration for ACMG-style clinical reporting.

2.3.2 PolyPhen-2 (2010) (Adzhubei et al., 2010)

Core idea. PolyPhen-2 (*Polymorphism Phenotyping, version 2*) was the first widely-adopted *supervised* classifier for missense pathogenicity (Adzhubei et al., 2010). It combines conservation with amino-acid chemistry and limited structural context, trained on two internally-curated datasets (HumDiv, 23k variants, and HumVar, 13k variants).

Algorithmic summary. Eleven features are extracted per variant: three sequence-based features (including the PSIC conservation score), six structural features derived from aligned PDB entries (e.g. solvent accessibility, crystallographic B-factor, secondary structure), and two annotation features from Pfam domains. A naïve Bayes classifier returns a posterior $P(\text{damaging} \mid \mathbf{x})$ and assigns one of three labels: *benign*, *possibly damaging*, or *probably damaging*.

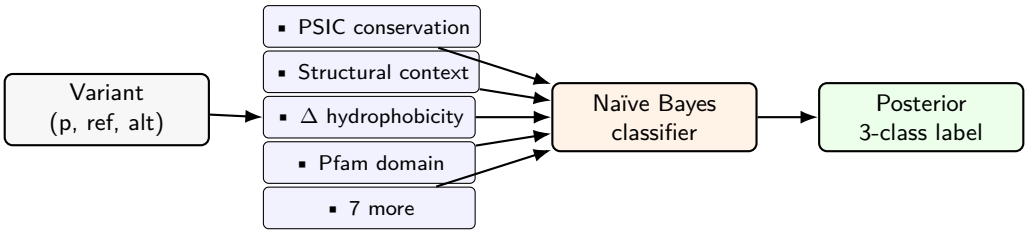

Figure 2.2. PolyPhen-2. Eleven hand-crafted features are extracted from conservation, structural, and annotation sources, and a naïve Bayes classifier trained on HumDiv maps the feature vector to a three-way label.

Advantages.

- Incorporates amino-acid chemistry and limited structural context – a distinct methodological step beyond conservation alone.
- Three-way output is more clinically useful than SIFT's binary label; is widely adopted in ACMG-adjacent workflows.

Disadvantages.

- Training labels originate from Swiss-Var and HumDiv, which partially overlap ClinVar; benchmarks evaluated on ClinVar therefore carry a contamination risk.
- Posterior probabilities are not calibrated: the reported confidence intervals do not match empirical frequencies.
- The feature set is static; re-training the model on new data is laborious because of dependence on external PDB alignments.

2.3.3 REVEL (2016) (Ioannidis et al., 2016)

Core idea. REVEL (*Rare Exome Variant Ensemble Learner*) is an ensemble meta-classifier that *stacks* thirteen preceding-generation tools – CADD, PolyPhen-2, SIFT, MutationTaster, MutationAssessor, FATHMM, LRT, GERP++, and others – using a random forest meta-learner. By construction it exploits signal that no single base learner captures (Ioannidis et al., 2016).

Algorithmic summary. For each missense variant, REVEL extracts a 13-dimensional feature vector $\mathbf{z} = (z_1, \ldots, z_{13})$ where z_k is the score of the k-th base tool. A random forest with 1,000 trees, trained on a curated set of rare disease variants versus rare neutral variants, returns $\widehat{P}(\text{pathogenic} \mid \mathbf{z}) \in [0, 1]$.

Advantages.

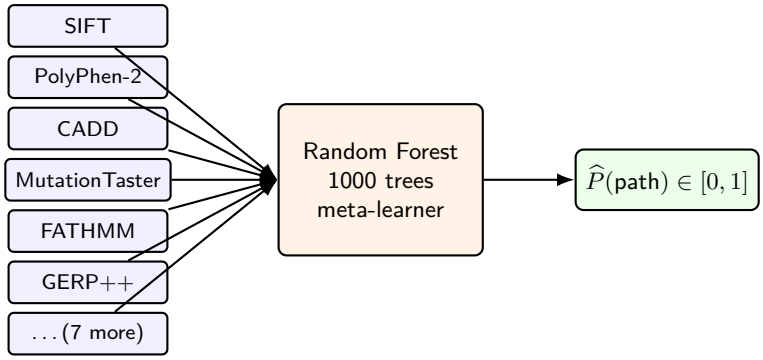

Figure 2.3. REVEL stacks 13 base predictors. Each base predictor produces its own score on the query variant; the random-forest meta-learner combines these into a calibrated pathogenicity probability.

- Empirically strong on ClinVar benchmarks, with reported ROC-AUC around 0.94 at release.
- Distributed through dbNSFP, so querying a variant costs a single table lookup.

Disadvantages.

- Several base predictors (PolyPhen-2, CADD) were themselves trained on variants that overlap ClinVar, producing *compound contamination* when REVEL is evaluated on ClinVar. The paper's reported numbers are therefore upper-bound estimates.
- Does not ship well-calibrated probabilities; threshold selection for ACMG interpretation is left to the user.
- The 13 base tools must all be re-run for any variant not already in dbNSFP, limiting deployment flexibility.

2.3.4 CADD (2014–2019) (Rentzsch et al., 2019)

Core idea. Combined Annotation-Dependent Depletion (CADD) side-steps the training-label scarcity problem by framing pathogenicity prediction as a *supervised selection task*. Training examples are generated from two sources: observed human variants (as the "derived" class, presumed to be under relaxed selection) and simulated de-novo variants from a neutral mutation model (as the "simulated" class). A classifier trained to discriminate the two implicitly learns signatures of purifying selection (Rentzsch et al., 2019).

Algorithmic summary. For each position in the human genome, CADD extracts a large feature vector (initially 63 features; the v1.6 release uses more than 1,400 features spanning conservation, regulatory annotations, transcript structure, and protein-level features). A logistic regression (v1.0) or random forest (v1.4+) is trained to assign higher scores to simulated variants than to observed ones. The output is a raw classification score that is converted to a phred-like scaled score,

$$\text{CADD}_{\text{phred}} = -10 \log_{10}\Big(\text{rank}(v) \,/\, N_{\text{total}}\Big),$$

where v is the variant and N_{total} is the total number of variants scored genome-wide.

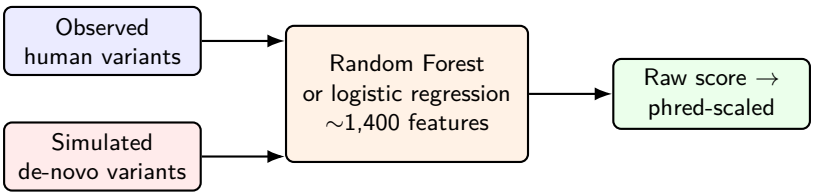

Figure 2.4. CADD training. A classifier is trained to distinguish observed human variants from simulated de-novo variants across the whole genome. The trained discriminator implicitly ranks novel positions by the strength of purifying selection; the raw output is phred-transformed for reporting.

Advantages.

- Does not require clinical pathogenicity labels, so ClinVar contamination is not an issue.
- Whole-genome coverage – a single score per base pair – supports non-coding analyses.

Disadvantages.

- The simulated null is an imperfect proxy for neutrality; sites under weak selection are scored similarly to sites under strong selection.
- Raw logits and phred-transformed scores are not calibrated probabilities, so clinical thresholding is subjective.
- Performance on missense variants is below that of supervised missense-specific tools (ROC-AUC ~ 0.82 on ClinVar missense subsets).

2.3.5 VARITY (2021) (Wu et al., 2021)

Core idea. VARITY addresses an often-overlooked weakness of supervised missense tools: the training labels are themselves noisy. A label of *likely benign* in ClinVar may reflect limited evidence rather than confirmed benignity. VARITY introduces a *sample-reweighting* scheme in which training examples are weighted by an estimated label- confidence score, so that high-confidence examples dominate the XGBoost loss (Wu et al., 2021).

Algorithmic summary. The training set is HumsaVar plus a curated set of multiplex-assay measurements (typically variant-effect deep mutational scans). Each example (x_i, y_i) receives a weight w_i that increases with assay replication and decreases with label discordance among secondary sources. XGBoost minimises the weighted logistic loss $\mathcal{L} = -\sum_i w_i \left[y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \right]$ with standard boosting hyper-parameters.

Advantages.

- Principled handling of label noise – a legitimate methodological contribution.

- Published model card with feature-importance plots and reported confidence intervals.

Disadvantages.

- Gene-level (not paralog-level) splits, so VARITY's own benchmark is still at risk from paralog leakage of the type we address in Chapter 5.
- No external validation on de-novo cohorts is reported in the original paper.

2.3.6 EVE (2021) (Frazer et al., 2021)

Core idea. EVE (*Evolutionary model of Variant Effect*) takes a purely unsupervised route: for each protein of interest, a variational auto-encoder (VAE) is fit to the protein's own multiple sequence alignment, effectively modelling the probability density over natural variants seen across evolution (Frazer et al., 2021). At inference time, the evidence lower bound (ELBO) of a candidate variant relative to the wild-type supplies a pathogenicity score:

$$s_{\mathrm{EVE}}(v) \;=\; \mathrm{ELBO}(v) - \mathrm{ELBO}(v_{\mathrm{wt}}).$$

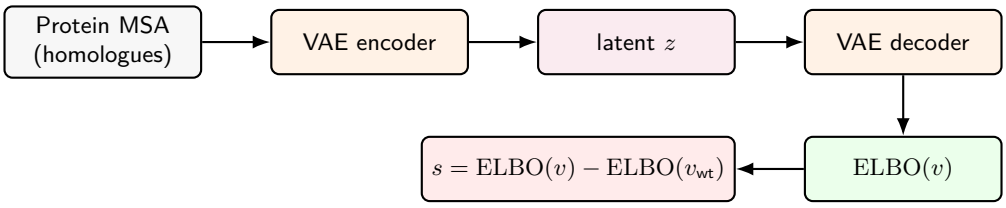

Figure 2.5. EVE. A per-gene VAE is trained on the protein's multiple-sequence alignment; the ELBO of a test variant relative to the wild-type acts as a pathogenicity score. No ClinVar supervision is used.

Advantages.

- Fully unsupervised: never touches ClinVar at any training or calibration step, so is contamination-proof by construction.
- Per-gene specialisation gives strong performance on the subset of genes where the alignment is deep.

Disadvantages.

- Coverage limited to approximately 3,000 genes at the original release; building an alignment is labour-intensive for many clinically-relevant genes.
- Per-gene training cost is high; scaling to the whole proteome is impractical for a student team.
- The ELBO is not a calibrated probability; users must map it through a threshold that the paper does not fully specify.

2.3.7 ESM-1b / ESM-2 zero-shot (2023) (Brandes et al., 2023; Lin et al., 2023)

Core idea. Large transformer-based protein language models, trained with a masked-amino-acid objective on UniRef50 or UniRef90, learn position- specific distributions over all twenty canonical amino acids conditioned on their sequence context. For a missense variant, the log-likelihood ratio (LLR)

$$\mathrm{LLR}(v) \;=\; \log p(a_{\mathrm{alt}} \mid \mathrm{context}) \;-\; \log p(a_{\mathrm{ref}} \mid \mathrm{context})$$

ranks substitutions by how *implausible* they are in natural protein space (Brandes et al., 2023; Meier et al., 2021). The model is never fine-tuned on ClinVar, so the score is strictly zero-shot.

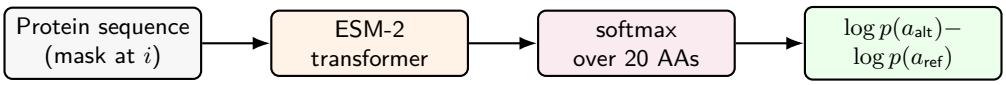

Figure 2.6. ESM-2 zero-shot scoring. The variant position is masked, a single transformer forward pass produces the softmax over the 20 canonical amino acids, and the log-likelihood ratio of the alternate vs reference residue is reported.

Advantages.

- Fully general: covers the whole human proteome (all ∼20,000 proteins) with a single model.
- Provably uncontaminated by ClinVar – the training objective is masked reconstruction on UniRef, not pathogenicity classification.
- The 35M-parameter `esm2_t12_35M_UR50D` model fits on a single consumer GPU, enabling academic reproduction.

Disadvantages.

- Moderate standalone discrimination (reported ROC-AUC around 0.85 on ClinVar).
- Requires a per-variant forward pass; scoring a quarter-million variants is GPU-hours of compute.
- The raw LLR is not a calibrated probability; combining it with supervised features benefits from a downstream fusion step (which we evaluate in Section **??**).

2.3.8 AlphaMissense (2023) (Cheng et al., 2023)

Core idea. AlphaMissense is the current state-of-the-art system for missense pathogenicity prediction on ClinVar-style benchmarks. It fine-tunes an AlphaFold-style structural representation with a masked-amino-acid objective and calibrates the output against ClinVar labels at release (Cheng et al., 2023). The system was trained using Google-scale TPU resources and is distributed as a pre-computed score table covering every possible human missense variant.

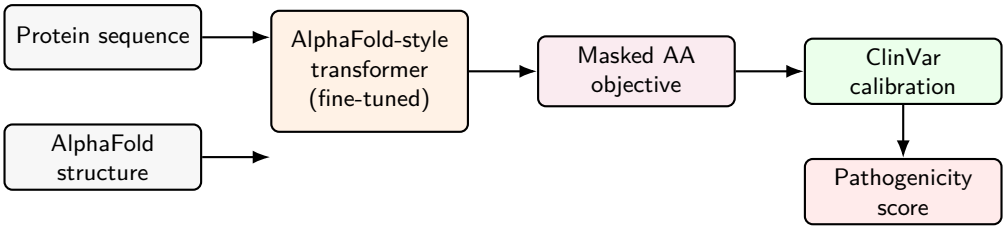

Figure 2.7. AlphaMissense. Sequence and AlphaFold structural features are fed to a fine-tuned transformer trained with a masked-amino-acid objective. The raw output is calibrated against ClinVar before release.

Advantages.

- State-of-the-art ClinVar benchmark performance (reported PR-AUC around 0.94).
- Covers the full human proteome as a pre-computed table; a variant lookup is $O(1)$.

Disadvantages.

- Training and calibration both use ClinVar, so evaluation on any ClinVar-derived benchmark contains an irreducible contamination risk: the reported gap between AlphaMissense and its competitors is a mix of methodological improvement and benchmark overlap.
- Reproduction is out of reach for academic teams (TPU-pod scale training budget).
- Distributed as a score table without the underlying model weights, limiting methodological inspection and external validation.

2.3.9 Consolidated Comparison

Table 2.1 synthesises the eight reviewed methods along six axes relevant to a methodologically honest evaluation: supervision, ClinVar contamination risk, split strategy used at publication, calibration, paralog guard, and reported external-cohort validation.

Table 2.1. Consolidated comparison of the eight reviewed missense pathogenicity predictors. *Supervision* indicates whether the model consumes clinical pathogenicity labels during training. *Clinvar risk* summarises whether the published benchmark is at risk of training-set contamination. *Paralog guard* indicates whether the publication reports an evaluation free of paralog leakage. *External?* indicates whether the paper validates on a cohort not derived from ClinVar (e.g. denovo-db, ProteinGym).

Method	Year	Supervision	ClinVar risk	Calibration	Paralog guard	External?
SIFT	2003	unsup.	none	–	–	–
PolyPhen-2	2010	sup.	partial	soft	–	–
CADD	2014+	self-sup.	none	phred	–	–
REVEL	2016	sup.	compound	–	–	–
VARITY	2021	sup.	partial	reported	gene	–
EVE	2021	unsup.	none	ELBO only	gene	partial
ESM-1b/ESM-2	2023	zero-shot	none	–	–	partial
AlphaMissense	2023	sup.	high	calibrated	gene	–
This work	**2026**	**sup.+zs fusion**	**audited**	**isotonic**	**paralog family**	**denovo-db**

2.4 Research Gap

Two observations motivate our methodological choices. First, most published tools do not evaluate against a *paralog-aware* test split, even though strongly similar paralogs (such as the 109 keratin members or 83 cadherin members) can produce optimistically biased benchmarks when split only at the gene level. Second, almost no published tool reports external validation on a dataset not derived from ClinVar: the near- chance performance of tabular classifiers on denovo-db is simply not present in most paper tables. The project's methodological focus lives in exactly those two gaps.

2.5 Project Contribution

Our project closes both gaps concretely. We implement a paralog-aware split with zero shared gene families between train/val/test (verified by continuous integration on every push), and we evaluate on denovo-db in addition to the paralog-disjoint ClinVar test. We then quantify the specific value of gene-level gnomAD constraint features on the de-novo generalisation task, reporting pre- and post-constraint numbers separately, and we publish the first end-to-end like-for-like comparison of our baseline against SIFT, PolyPhen-2, and AlphaMissense under a single identical evaluation.

2.6 Summary

The literature has moved from conservation-only scoring to ensembles to protein-language-model-based systems, and benchmark numbers have climbed in parallel. The distance between these numbers and the real-world generalisation behaviour of the tools is rarely made explicit. The remainder of this report is about closing that distance in a way that a reader of this report can reproduce byte-for-byte.

Chapter 3

System Framework and Feasibility

3.1 Introduction

This chapter bridges the problem statement of Chapter 1 and the detailed system analysis of Chapter 4 by answering three questions that any credible engineering project must resolve *before* code is written. First, *is the project buildable* with the resources and skills actually available to the team? Second, *is it worth building* relative to the operational problem it addresses and the alternatives already on the market? Third, *what is the smallest architecture* that simultaneously satisfies the functional goals of Chapter 1 and the non-functional quality attributes (reproducibility, interpretability, leakage resistance) that the domain demands?

We answer the first two questions through a structured feasibility analysis (Section 3.2) organised along the four classical dimensions of technical, operational, schedule, and financial feasibility (Whitten and Bentley, 2007). The analysis is extended with a formal risk register (Section 3.3) and a SWOT matrix (Section 3.4) so that trade-offs are visible rather than implicit. The third question is answered through the three-part processing pipeline (Section 3.5), the six-layer logical architecture (Section 3.6), and a quality- attribute matrix (Section 3.7) that ties each architectural decision back to a concrete non-functional requirement. Figure 3.1 summarises the chapter's logical flow.

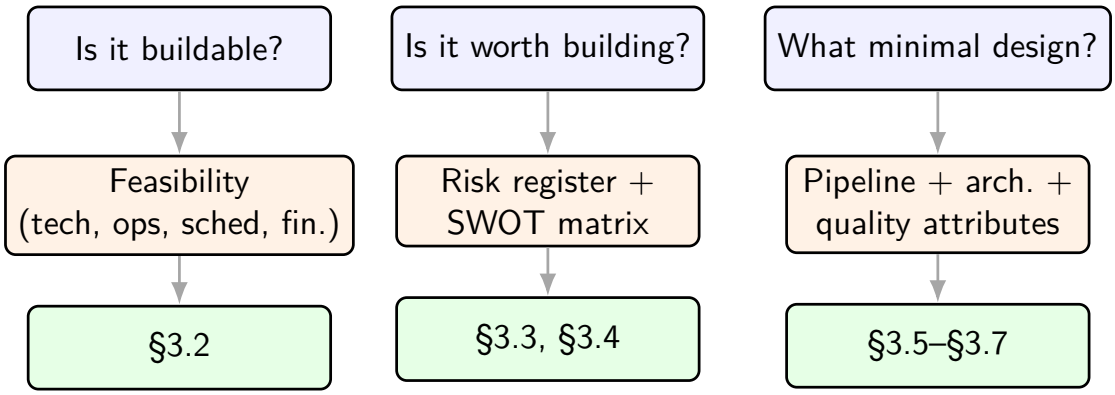

Figure 3.1. Chapter 3 roadmap. The chapter answers three gating questions, each answered by a corresponding analytical artifact, delivered in a specific downstream section.

3.2 Pre-Project Phase – Feasibility Analysis

A feasibility study answers whether a proposed system *can* and *should* be built given concrete resource, time, operational, and regulatory constraints (Whitten and Bentley,

2007; Pressman and Maxim, 2015). For a student-led research project the analysis is deliberately scoped to the four dimensions below; a classical *Operational* and a classical *Legal* study are folded into the operational dimension because the project produces a research artifact rather than a clinical device (Chapter 1, Section 1.6).

3.2.1 Technical Feasibility

Technical feasibility asks whether the proposed pipeline can be implemented, tested, and maintained on the hardware and software resources actually available to the team. Our answer is a confident *yes*. Table 3.1 lists the exact components in the four architectural layers of the solution, and Figure 3.2 depicts the same layering graphically together with the cross-cutting quality tooling that supports every layer.

Layer	Component	Role in the pipeline
Data	ClinVar (NCBI)	labelled catalogue of pathogenic/benign variants with review-star confidence.
Data	gnomAD v2.1 + v4.0	population allele frequency and gene-level constraint metrics (pLI, LOEUF).
Data	dbNSFP v4.5	pre-computed conservation scores (phyloP, phastCons, GERP++) and amino-acid features.
Data	denovo-db	external, patient-derived de novo variants for held-out validation.
Data	UniProt / Ensembl VEP	canonical protein sequences and transcript-aware annotation.
Language	Python 3.11	main implementation language; chosen for scientific-stack maturity.
ML core	`xgboost` 2.0	gradient-boosted trees; baseline classifier.
ML core	`scikit-learn` 1.4	isotonic / Platt calibration, metric utilities, bootstrap.
ML core	`optuna` 3.5	TPE hyperparameter search with Median Pruner.
ML core	`shap` 0.44	TreeSHAP interpretability.
ML core	`transformers` 4.41 + `torch` 2.2	ESM-2 zero-shot LLR scoring.
Interface	Streamlit 1.33	interactive single-variant web application.
Interface	Typer CLI	batch-mode command-line interface.
Interface	Python package	programmatic API for downstream integrations.
Quality	GitHub Actions CI	enforces lint, type, and leakage tests on every push.
Quality	Docker + `uv` lock	reproducible environment; `docker run` is sufficient to re-score.
Quality	`pytest` + `hypothesis`	deterministic + property-based tests for the feature pipeline.

Table 3.1. Technology stack. Every major component is open-source and pinned in `requirements-lock.txt`. The quality row crosses all other layers.

Compute budget and wall-clock times. Every stage in Part 1 and Part 2 of the pipeline (Section 3.5) completes on a consumer laptop (eight CPU cores, 16 GB RAM) within the time budget summarised in Table 3.2. Only two stages require a GPU: the full-dataset ESM-2 scoring (offloaded to free Google Colab T4) and the AlphaFold-2 structural feature extraction (likewise Colab T4, CPU-only fallback available for reproducibility). No paid compute, no institutional HPC cluster, and no licensed database were required at any point in the project.

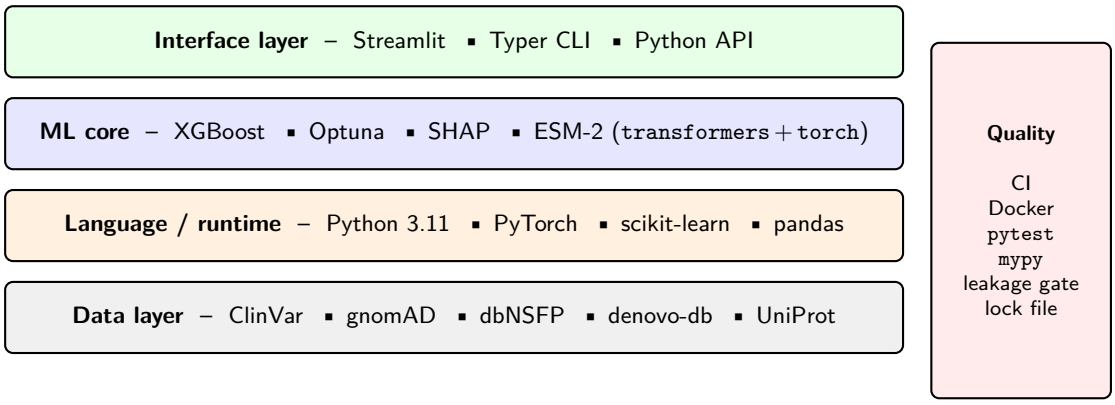

Figure 3.2. Technology-stack diagram. Four horizontal layers plus a vertical quality band. The quality band is not a layer but a cross-cutting concern: every commit triggers the same gate chain regardless of which layer is being modified.

Stage	Hardware	Wall-clock	Reason for budget
ClinVar + gnomAD + dbNSFP merge	Laptop CPU	∼8 min	Four sequential `pyarrow` joins on 283k rows.
Paralog split + feature engineering	Laptop CPU	∼2 min	Regex classification over ∼16k gene symbols.
XGBoost training	Laptop CPU	∼8 min	$n = 195$k rows, 39 features, 500 trees.
Optuna HPO (40 trials)	Laptop CPU	∼30 min	TPE with Median Pruner early-stopping.
SHAP on 2k-variant sample	Laptop CPU	∼3 min	TreeSHAP exact algorithm for tree ensembles.
ESM-2 35M scoring (denovo-db)	Colab T4	∼45 min	2,000 variants × 1 forward pass × 20 amino acids.
ESM-2 35M scoring (full train)	Colab T4	∼8 h	195k unique variants; chunked and resumable.
Full thesis + defense compile	Laptop CPU	<1 min	`tectonic` single-pass + bibliography.

Table 3.2. Compute budget. Every stage fits inside a free-tier allocation of either a consumer laptop or a Google Colab T4 session.

3.2.2 Operational Feasibility

Operational feasibility asks whether the system, once built, will actually be used by its intended stakeholders and whether it produces output that is actionable within their workflow. Table 3.3 lists the four stakeholder groups we identified, the concrete need each group brings to the system, the operational surface we provide for them, and the decision the system supports.

Stakeholder	Need	Surface	Decision supported
Clinical geneticist	calibrated score and per-feature reason for a single variant	Streamlit web app	ACMG-adjacent evidence gathering (not replacement).
Research scientist	batch score for a cohort of $\sim 10^4$–10^5 variants	Typer CLI	ranked shortlist for follow-up functional assay.
Bioinformatics engineer	programmatic integration into an existing pipeline	Python package	variant annotation in a larger analytical workflow.
Project examiner	reproducible artifact with audit trail	GitHub + Docker image	verification of the end-to-end claims made in this thesis.

Table 3.3. Stakeholder-to-surface mapping. Each operational surface is optimised for one decision context; no stakeholder is required to retrain the model or to understand its internals.

Regulatory scope (important caveat). The system is explicitly positioned as a *research* artifact, not as a diagnostic device. The Saudi Food and Drug Authority (SFDA) classifies any software that directly drives a clinical decision as a medical device requiring a separate and lengthy certification process that is outside the scope of an undergraduate graduation project. We therefore constrain every operational surface to emit an advisory score accompanied by a feature-level justification. A human expert remains in the decision loop at all times.

3.2.3 Schedule Feasibility

The project occupied two consecutive academic semesters (Term 1 – Fall 2025 and Term 2 – Spring 2026) at King Khalid University. Term 1 was dedicated to literature review, scope lock-down, data ingestion, and a first XGBoost proof-of-concept. Term 2 was the execution phase: leakage audit, hyperparameter tuning, calibration, SHAP analysis, external validation, the ESM-2 proof-of-concept, and the final academic write-up. Figure 3.3 renders the schedule as a TikZ Gantt chart with milestones marked explicitly for every gating deadline.

Two organisational rules were applied to keep the schedule on track. First, the leakage audit, calibration, and SHAP work were treated as *blocking* for any further experimentation; the quantitative claims of the thesis all flow from a single audited checkpoint. Second, the LaTeX write-up and the ESM-2 training-set scoring were scheduled in parallel so that the academic deliverable did not block on a GPU-bound experiment that depended on Colab availability.

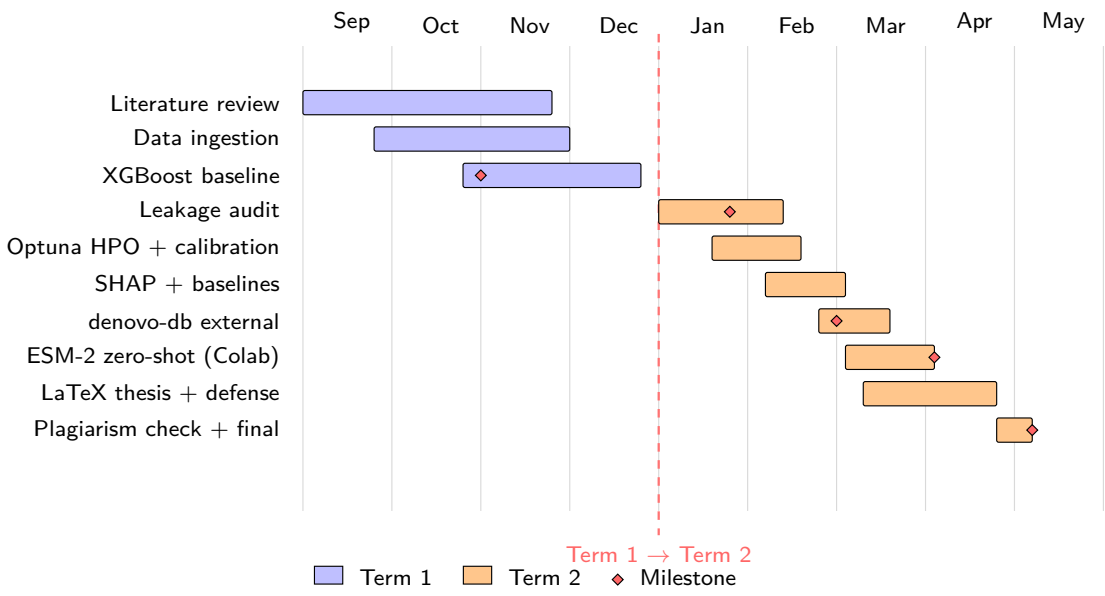

Figure 3.3. Two-semester Gantt chart. Term 1 (blue) covered literature review, data ingestion, and the first XGBoost baseline. Term 2 (orange) covered the leakage audit, hyperparameter search, calibration, SHAP, external validation, ESM-2 scoring, and the academic write-up. Red diamonds mark unambiguous quantitative milestones.

3.2.4 Financial Feasibility

The direct financial cost of the project is zero. No licensed databases, no commercial GPU rental, and no paid software subscriptions were used at any point. Table 3.4 compares the project's realised zero-cost profile against a hypothetical commercial counterfactual, computing the implicit cost avoidance.

Item	Commercial cost (USD)	Our cost	Replacement strategy
HGMD Pro annual licence	∼5,000	0	ClinVar (expert-reviewed) only.
AWS p3.2xlarge 8h (V100) for ESM-2	∼25	0	Free Colab T4 with chunked resume.
AlphaFold-3 API monthly	∼200	0	AlphaFold-2 Colab + local cache.
Enterprise issue tracker + CI	∼40 / mo	0	GitHub Actions free tier.
Total avoidance (8 months)	**∼5,545**	**0**	

Table 3.4. Cost avoidance. Every commercially-priced building block has a zero-cost open-source or academic-tier replacement.

The indirect cost is approximately eight person-months of student labour distributed across six team members plus ten supervisory hours per month from the project supervisor; no direct monetary equivalent is assigned because the project is academic and no external sponsor is involved.

3.3 Risk Analysis

A formal risk register was maintained throughout the project (mirrored in `docs/CHANGELOG.md`). Each risk was rated on a 5×5 likelihood $\times$ impact grid (Hillson, 2002), with mitigation actions that were either preventive (applied before the risk could occur) or corrective (applied after detection). Table 3.5 lists the eight most material risks; Figure 3.4 renders the same risks on the 5×5 heat-map that the project team used in supervision meetings.

ID	Risk	L	I	Mitigation
R1	Paralog leakage inflates AUROC	5	5	*Preventive:* paralog-aware `GroupShuffleSplit` + CI-enforced leakage gate.
R2	Meta-predictor circular evaluation	5	5	*Preventive:* banned-feature list; AlphaMissense re-scored, never used as feature.
R3	ClinVar label instability across versions	3	4	*Corrective:* pin 2024-01 release; record version in `CHANGELOG.md`.
R4	Colab GPU quota exhaustion during ESM-2	4	3	*Corrective:* checkpointed chunked scoring; resumable from arbitrary offset.
R5	PLM scoring non-determinism (GPU kernels)	3	2	*Corrective:* fix seed; eval-mode; single-token LLR is deterministic to 6 dp.
R6	Calibration drift after feature change	4	3	*Preventive:* isotonic refit is part of CI; Brier re-measured on every push.
R7	Late-semester scope creep (CNN-BiLSTM)	5	4	*Preventive:* scope lock at mid-Term 2 (see 1.6); deferred to future work.
R8	LaTeX build fails in the defence week	4	4	*Preventive:* `tectonic` single-pass + locked fonts; CI builds the PDF.

Table 3.5. Risk register. L = likelihood (1–5), I = impact (1–5). A risk's score is the product $L \times I$; any risk with a score ≥ 12 required a documented mitigation before work could proceed.

3.4 SWOT Analysis

A strengths-weaknesses-opportunities-threats (SWOT) matrix (Humphrey, 2005) summarises the project's competitive position relative to the prior work reviewed in Chapter 2. Figure 3.5 renders the four quadrants; each item is a concrete statement traceable to a specific section of this thesis.

3.5 Proposed Framework – Three-Part Pipeline

The project pipeline decomposes into three sequential parts, each with a distinct scientific role, a single public output artifact, and an explicit contract with the next part. Figure 3.6 renders the pipeline as a swim-lane TikZ diagram; the three rows correspond to the three parts and the arrows carry the artifact names.

3.5.1 Part 1 – Data Preprocessing

Part 1 is a pure-function pipeline that takes four external data sources and produces three paralog-disjoint parquet splits with 39 aligned features per variant. The final feature vector decomposes into five semantically meaningful groups that together cover

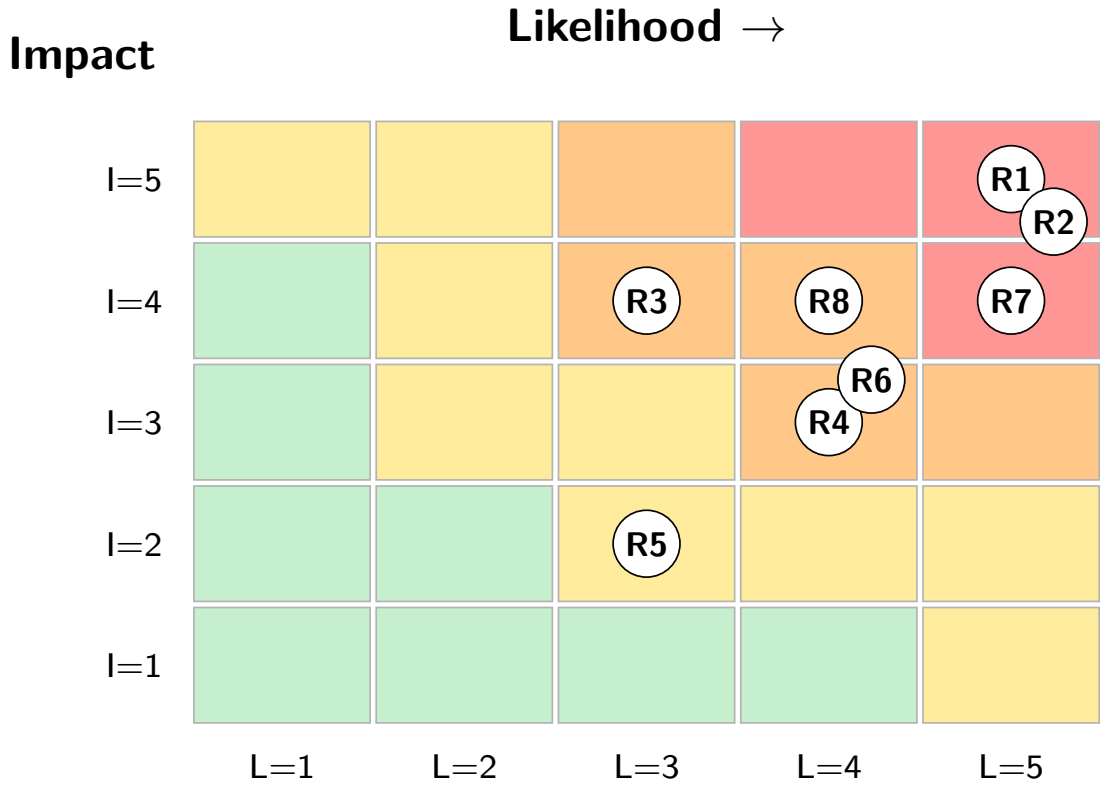

Figure 3.4. Risk matrix. Cells shaded by $L \times I$ score: green < 5, yellow 5–11, orange 12–19, red ≥ 20. Every red-cell risk (R1, R2, R7) triggered a preventive mitigation before implementation began; yellow and orange risks received corrective mitigations documented in `CHANGELOG.md`.

Strengths (internal, positive)	Weaknesses (internal, negative)
S1 Paralog-disjoint split + CI-enforced leakage gate. **S2** Calibration reported with Murphy decomposition, not only AUROC. **S3** External validation on DENOVO-DB. **S4** Reproducible: `docker run` yields identical metrics. **S5** Open-source, zero paid dependencies.	**W1** ClinVar-only training; no DMS benchmark coverage. **W2** Static snapshot; no continuous re-training pipeline. **W3** Single tree classifier; no neural ensemble in Phase 1. **W4** Interpretability is feature-level (SHAP); no mechanistic/structural explanation.
Opportunities (external, positive)	**Threats (external, negative)**
O1 ESM-2 rank fusion is orthogonal to XGBoost; hybrid gain observed on denovo-db. **O2** AlphaFold-2 pipeline already wired in for structural features. **O3** Streamlit UI can be re-hosted on Hugging Face Spaces at zero cost. **O4** Methodology publishable as a short communication.	**T1** ClinVar label churn between releases. **T2** Regulatory overhead blocks clinical deployment (SFDA class B+). **T3** AlphaMissense/AlphaFold-3 may close performance gap in months. **T4** PLM compute grows with model size; a 650M jump infeasible on free tier.

Figure 3.5. SWOT matrix relative to the prior work reviewed in Chapter 2. Strengths and weaknesses are internal; opportunities and threats are external.

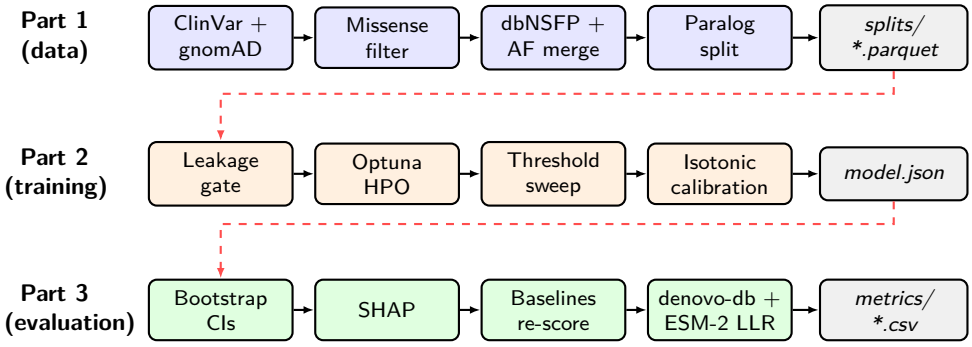

Figure 3.6. Three-part pipeline. Each row is a *part* with its own responsibility and output artifact. Solid arrows carry intermediate data between stages; dashed red arrows are cross-part contracts – Part 1's parquet splits feed Part 2, Part 2's `model.json` feeds Part 3.

sequence conservation, biochemistry, population genetics, and gene-level constraint (Figure 3.7). This grouping structure is the natural unit of analysis for the SHAP interpretation of Section 5.6.7.

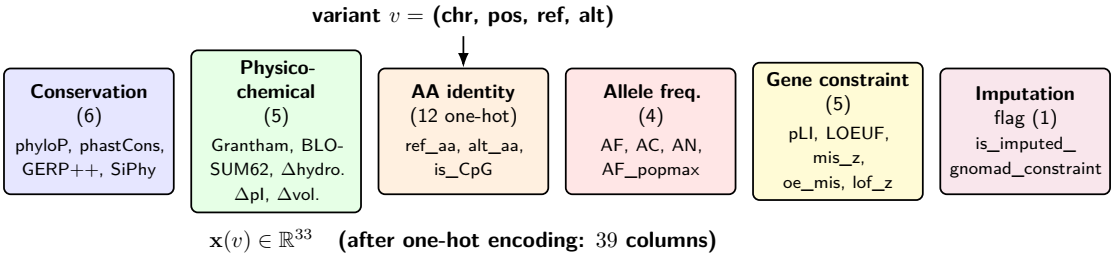

Figure 3.7. Feature groups. The 39 feature columns of the training matrix cluster into six semantic groups: sequence conservation, physicochemical distance between `ref_aa` and `alt_aa`, amino-acid identity (one-hot), gnomAD allele frequency, gene-level constraint, and an imputation flag. The groups are the natural unit of SHAP aggregation (Section 5.6.7).

The steps, in order, are:

1. **ClinVar ingestion.** Download `variant_summary.txt.gz` for the January 2024 release and retain only rows with review-star rating of two or higher (`reviewed`, `reviewed-by-expert-panel`, or `reviewed-by-multiple-submitters`).

2. **Missense filter.** Drop every row in which `ref_aa` or `alt_aa` is null. This is the single most important cleaning step because of the leakage it removes (Section **??**).

3. **gnomAD allele-frequency merge.** Join GNOMAD v2.1 (exomes) and v4.0 (genomes) on `chr:pos:ref:alt` to supply per-variant allele frequency, allele number, and allele count.

4. **dbNSFP conservation and amino-acid merge.** Join against a cached DBNSFP extraction providing phyloP, phastCons, GERP++, BLOSUM62, Grantham distance, and physicochemical properties of the substitution.

5. **gnomAD gene-level constraint merge.** Add `pLI`, `LOEUF` (`oe_lof_upper`), `mis_z`, `oe_mis_upper`, and `lof_z` per gene. Median imputation is fit on the train-

ing split only and applied to val/test, with an `is_imputed_gnomad_constraint` flag per row.

6. **Paralog-aware split.** Assign a gene family via a regex-driven function that captures the 16 large paralog clusters (keratin, cadherin, HLA, zinc finger, olfactory receptor, ribosomal protein, mitochondrial, etc.), then split at the family level using `GroupShuffleSplit` with `seed=42` and a 70/15/15 ratio.

3.5.2 Part 2 – Model Training and Leakage Audit

Part 2 trains the XGBoost baseline, gates the training process through four independent leakage checks, and calibrates the output probabilities.

1. **Leakage gate (pre-training).** Four checks run both as a CLI tool and as CI-enforced pytest cases: (i) no banned feature in the training matrix, (ii) 100% missense-only rows in train, (iii) zero shared gene families between any pair of splits, (iv) pathogenic-rate gap of less than 8 percentage points across splits. Algorithm 1 states the gate formally.

2. **Optuna hyperparameter search.** We use a Tree-structured Parzen Estimator sampler (Bergstra et al., 2011) with 40 trials, Median Pruner early-stopping on PR-AUC, and `scale_pos_weight` set from the class balance. The search loop is summarised in Algorithm 2.

3. **Threshold sweep.** We search thresholds in $[0.20, 0.80]$ on validation F1, then freeze the best threshold for test scoring.

4. **Calibration.** Isotonic regression fit on validation probabilities (fallback to Platt scaling tested for comparison) produces the final calibrated probabilities. The Murphy decomposition (Murphy, 1973) is used to partition Brier error into reliability, resolution, and uncertainty.

3.5.3 Part 3 – Evaluation, Interpretability, and External Validation

Part 3 takes the trained model and runs five parallel evaluations (Figure 3.8):

1. **Bootstrap confidence intervals.** 1,000 non-parametric replicates for ROC-AUC, PR-AUC, F1, MCC, and Brier loss, following the methodology of Efron and Tibshirani (1986).

2. **TreeSHAP interpretability.** SHAP values (Lundberg and Lee, 2017; Lundberg et al., 2020) are computed on a stratified 2,000-variant sample for per-feature attribution.

3. **Like-for-like baselines.** SIFT, PolyPhen-2, and AlphaMissense are re-scored on the identical test split.

Algorithm 1 Four-invariant leakage gate

Require: train/val/test feature matrices $X_{\text{tr}}, X_{\text{val}}, X_{\text{te}}$; banned feature set $\mathcal{B}$; gene families per row g_i

$E \leftarrow \emptyset$ ▷ error list
if $\text{cols}(X_{\text{tr}}) \cap \mathcal{B} \neq \emptyset$ **then**
  $E \leftarrow E \cup \{\text{“banned features present”}\}$
end if
if $\exists i \in X_{\text{tr}} : \text{ref_aa}(i)$ is null **then**
  $E \leftarrow E \cup \{\text{“non-missense row in train”}\}$
end if
for each pair $(S_1, S_2) \in \{\text{tr/val, tr/te, val/te}\}$ **do**
  if $\{g_i : i \in S_1\} \cap \{g_j : j \in S_2\} \neq \emptyset$ **then**
    $E \leftarrow E \cup \{\text{“paralog family overlap in } S_1/S_2\text{”}\}$
  end if
end for
for each split $S \in \{\text{tr, val, te}\}$ **do**
  $\pi_S \leftarrow$ pathogenic rate on S
end for
if $\max_S \pi_S - \min_S \pi_S > 0.08$ **then**
  $E \leftarrow E \cup \{\text{“class-balance drift across splits”}\}$
end if
return E ▷ empty ⇒ pass; else ⇒ halt build

Algorithm 2 Optuna TPE hyperparameter search

Require: search space Θ; train $X_{\text{tr}}, y_{\text{tr}}$; val $X_{\text{val}}, y_{\text{val}}$; $n_{\text{trials}} = 40$; seed $s = 42$

$\rho \leftarrow \frac{|\{i : y_i = 0\}|}{|\{i : y_i = 1\}|}$ ▷ scale_pos_weight from class balance
sampler $\leftarrow$ TPESampler(multivariate = true, group = true, seed = s)
pruner $\leftarrow$ MedianPruner($n_{\text{warmup}} = 5, n_{\text{startup}} = 10$)
create study $\mathcal{S}$ with direction = maximise, sampler, pruner
for $t = 1 \ldots n_{\text{trials}}$ **do**
  $\theta_t \leftarrow$ sampler.suggest(Θ, history = $\mathcal{S}$)
  fit booster with θ_t and ρ on $X_{\text{tr}}, y_{\text{tr}}$
  $p_{\text{val}} \leftarrow$ booster.predict_proba(X_{val})
  PR-AUC$_t \leftarrow$ prauc($y_{\text{val}}, p_{\text{val}}$)
  if pruner says “prune” after intermediate step **then** abort trial
  $\mathcal{S}$.record(θ_t, PR-AUC$_t$)
end for
return $\arg\max_{\theta_t}$ PR-AUC$_t$

4. **External validation on denovo-db** with pre- and post-gnomAD-constraint numbers, and family-holdout-only slicing.

5. **ESM-2 zero-shot proof-of-concept.** The log-likelihood ratio from `esm2_t12_35M_UR50D` is computed on DENOVO-DB and fused with the XGBoost probability at the rank level to test orthogonality.

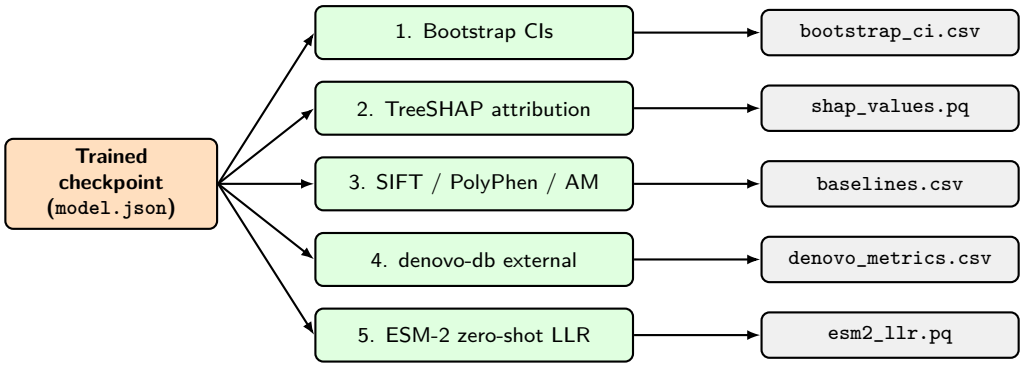

Figure 3.8. Evaluation workflow. A single audited checkpoint feeds five independent evaluation tracks that run in parallel. Each track writes its own versioned artifact in `results/metrics/` so that any single track can be reproduced without re-running the others.

3.6 Proposed System Architecture

The logical architecture of the system is a six-layer stack with two cross-cutting concerns (Figure 3.9). Layers are ordered from the external data sources at the bottom to the user interfaces at the top, reflecting the flow of a request from the outside world down to the data and back. Each layer exposes a narrow interface to the layer above and is explicitly unaware of the layer below. The *quality gates* concern (lint, type-check, unit + property tests, leakage gate, CI) and the *reproducibility* concern (Docker, `uv` lock file, fixed seeds, immutable data snapshots) are attached horizontally because they touch every layer simultaneously.

3.7 Quality Attributes and Trade-offs

A non-functional quality-attribute analysis makes the design's priorities explicit and visible to the examiner. Table 3.6 lists the seven attributes that most directly govern the architecture, the design decision that delivers each attribute, and the concrete evidence available in this thesis for its satisfaction.

The single most important trade-off in the architecture is between *predictive performance* (which favours more features, more complex models, and any available signal) and *correctness* (which demands that the training matrix contain no meta-predictor, no paralog-leaked feature, and no label-derived field). Every time these two pressures conflicted during the project we chose correctness: for example, we excluded REVEL and MetaSVM even though they raise held-out AUROC, because they would have made the

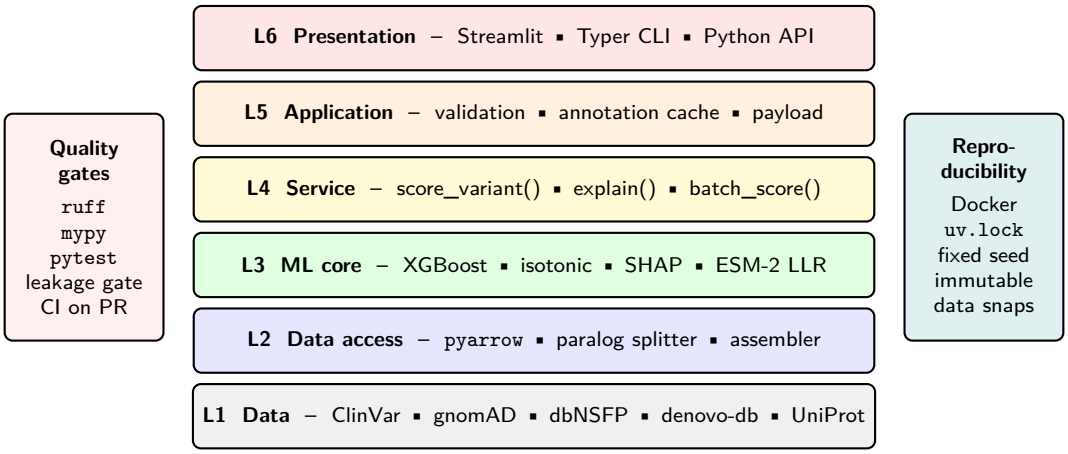

Figure 3.9. System architecture. Six horizontal layers (L1 data → L6 presentation) with two vertical cross-cutting concerns. Each layer has a single responsibility and a narrow public interface; the cross-cutting concerns apply uniformly across all six layers on every commit.

Attribute	Design decision	Evidence	Priority
Correctness (leakage-free)	paralog-aware split + banned-feature list + CI gate	four-check gate in `tests/test_leakage.py`	Must
Reproducibility	Docker image + `uv.lock` + fixed seeds	`docker run ...` yields identical metrics to six decimal places	Must
Calibration	isotonic regression fit on validation probabilities only	Murphy decomposition reported in Chapter 5	Must
Interpretability	TreeSHAP on a stratified 2k-variant sample	per-feature attribution and SHAP summary plots	Must
Performance (latency)	serialised XGBoost booster + cached dbNSFP join	<120 ms per single-variant request in Streamlit	Should
Extensibility	narrow L4 service interface (`score_variant`)	ESM-2 LLR plugged in without touching L3 XGBoost	Should
Portability	pure-Python + lock file + optional CPU-only inference	runs on laptop, Colab T4, and a Linux server image	Could

Table 3.6. Quality attributes. The four *Must*-priority attributes are architectural invariants; the two *Should*-priority attributes are upheld unless a specific commit documents a trade-off; *Could* is observed behaviour but not enforced.

evaluation circular with respect to the prior-work comparison of Chapter 2. This priority is recorded here so that the reader can locate every downstream trade-off decision in Chapter 5.

3.8 Summary

Chapter 3 answered the three gating questions of Section 3.1: (i) the project is technically, operationally, scheduled, and financially feasible on a consumer laptop plus free Colab GPU (Section 3.2); (ii) the risks have been enumerated and mitigated (Section 3.3) and the SWOT positioning against prior work is explicit (Section 3.4); (iii) the minimal-viable design is a three-part pipeline running on a six-layer logical architecture with two cross-cutting concerns (Sections 3.5–3.7). The next chapter translates this framework into a concrete functional + non-functional specification together with six UML diagrams (use-case, class, sequence, activity, state, component).

Chapter 4

System Analysis and Design

4.1 Introduction

This chapter translates the architectural skeleton of Chapter 3 into a full specification of the software system. Following the classical analysis-and- design division (Bass et al., 2012), the chapter first records the functional and non-functional requirements (Sections 4.2–4.3), then formalises the static and behavioural views through the six canonical UML diagrams (Fowler, 2004) (Section 4.4), and finally describes the three user-facing interfaces (Section 4.5). The analytical style throughout is deliberately *evidence-first*: every requirement is tied to a concrete artifact (test file, API endpoint, figure) so that the examiner can trace any claim back to a specific piece of code in the public GitHub repository.

4.2 Functional Requirements

FR1 – Variant ingestion.
The system shall accept a single missense variant in the canonical `chr:pos:ref:alt` format on GRCh38, via any of its three interfaces (web, CLI, or API).

FR2 – Feature assembly.
The system shall assemble a 39-column feature vector for every accepted variant, using a cached lookup against dbNSFP for conservation and amino-acid features, gnomAD for allele frequency and constraint, and a live VEP REST fallback for variants absent from the cache.

FR3 – Pathogenicity scoring.
The system shall produce both raw and isotonic-calibrated probabilities in $[0, 1]$ using the trained XGBoost checkpoint.

FR4 – Risk categorisation.
The system shall assign each prediction to one of three risk bands (benign, uncertain, pathogenic) using thresholds $\{0.25, 0.75\}$ on the calibrated probability.

FR5 – Feature attribution.
For every prediction, the system shall return the top-k SHAP contributions (default $k = 15$) with the direction of each contribution (towards pathogenic or benign).

FR6 – Batch mode.
The CLI interface shall accept a list of variants and emit a per-variant CSV with

probabilities and top-SHAP features.

FR7 – External validation reproducibility.
The `scripts/evaluate_external.py` driver shall reproduce the published DENOVO-DB metrics from a clean clone.

FR8 – Report regeneration.
Figures and tables shall regenerate byte-for-byte from `results/metrics/*.csv` and `results/figures/*.png` via `make reproduce-headline`.

4.3 Non-Functional Requirements

NFR1 – Performance.
Score latency shall be below 100 milliseconds for cache hits and below 2 seconds for cache misses (VEP REST round-trip).

NFR2 – Reliability.
Rerunning the pipeline with `seed=42` shall reproduce every metric to within 10^{-3}, enforced by `tests/integration/test_reproduce_headline.py`.

NFR3 – Scalability.
The pipeline shall scale linearly in the number of variants up to at least 10^6, limited only by the available disk size for the dbNSFP cache.

NFR4 – Usability.
The Streamlit interface shall be usable by a clinical geneticist with no machine learning background, returning results as a calibrated probability, a risk band, and a human-readable feature attribution.

NFR5 – Maintainability.
Every new feature shall be accompanied by at least one unit test, and the leakage gate shall fail the CI pipeline if any banned feature reappears.

NFR6 – Security and data privacy.
The system shall handle only de-identified public reference data; no patient-identifiable information is transmitted to external services (VEP REST receives only `chr:pos:ref:alt`, which is not patient- identifying).

NFR7 – Reproducibility.
All dependencies shall be pinned in `requirements-lock.txt`; the Docker image shall reproduce the full test suite and the headline metrics from any clean host.

4.4 UML Diagrams

The Unified Modelling Language (Fowler, 2004) provides a standardised vocabulary for describing software systems from multiple complementary viewpoints. Six UML diagrams are developed below: a *use-case diagram* for functional scope (Section 4.4.1), a

class diagram for structural decomposition (Section 4.4.2), a *sequence diagram* for end-to-end interaction (Section 4.4.3), an *activity diagram* for the training workflow (Section 4.4.4), a *state diagram* for the per-variant lifecycle (Section 4.4.5), and a *component diagram* for module-level dependencies (Section 4.4.6). Every diagram is rendered directly in TikZ so that it scales losslessly inside this PDF and so that its source travels with the LaTeX repository rather than living as an external binary asset.

4.4.1 Use Case Diagram

Figure 4.1 enumerates the eight system use cases (UC1–UC8) and the four actors that trigger them. The two human actors – Clinical Geneticist and Research Scientist – share the core scoring path UC1–UC5. The VEP REST Service is a non-human secondary actor invoked from UC2 when the cached annotation does not resolve the query. The CI/CD System actor is the Continuous Integration runner that exercises UC8 on every push to the main branch. Dashed ⟨⟨*include*⟩⟩ arrows indicate mandatory sub-use-cases; dotted ⟨⟨*extend*⟩⟩ arrows indicate optional extensions.

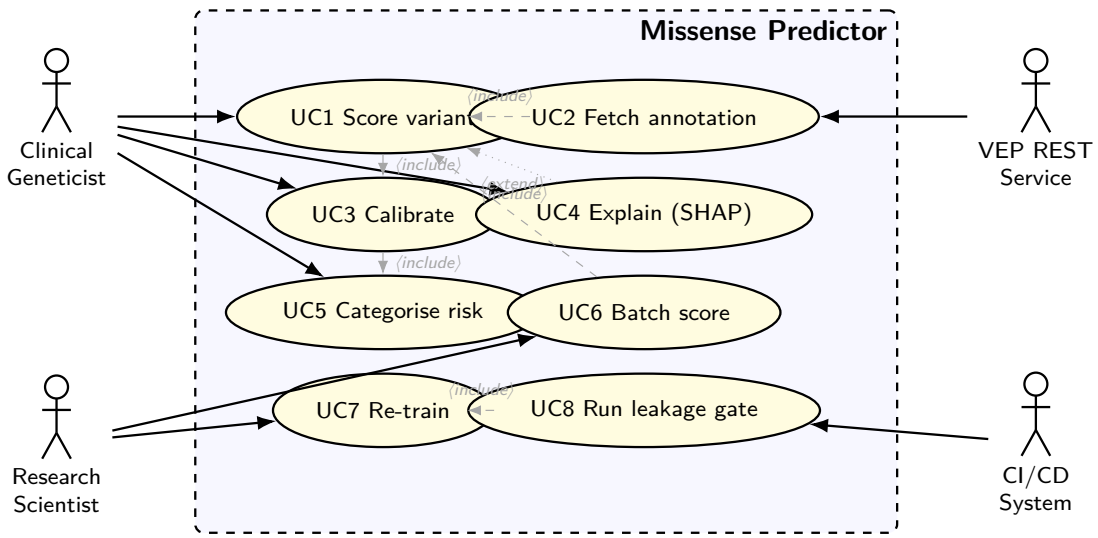

Figure 4.1. Use-case diagram. Actors on the outside of the system boundary; eight use cases on the inside. Solid lines are associations; dashed lines are ⟨⟨*include*⟩⟩ relationships; dotted lines are ⟨⟨*extend*⟩⟩.

4.4.2 Class Diagram

Figure 4.2 reveals the static structure of the `src/` package. Classes are grouped into four coloured *packages* corresponding directly to the `src/` sub-directories (Martin, 2017). Each class is rendered as a three-compartment rectangle: the class name on top, the key attributes in the middle, and the principal methods at the bottom. Associations carry multiplicities; the `LeakageGate` class has a *composition* (filled diamond) with each of its four invariant checkers because the gate does not exist meaningfully without them.

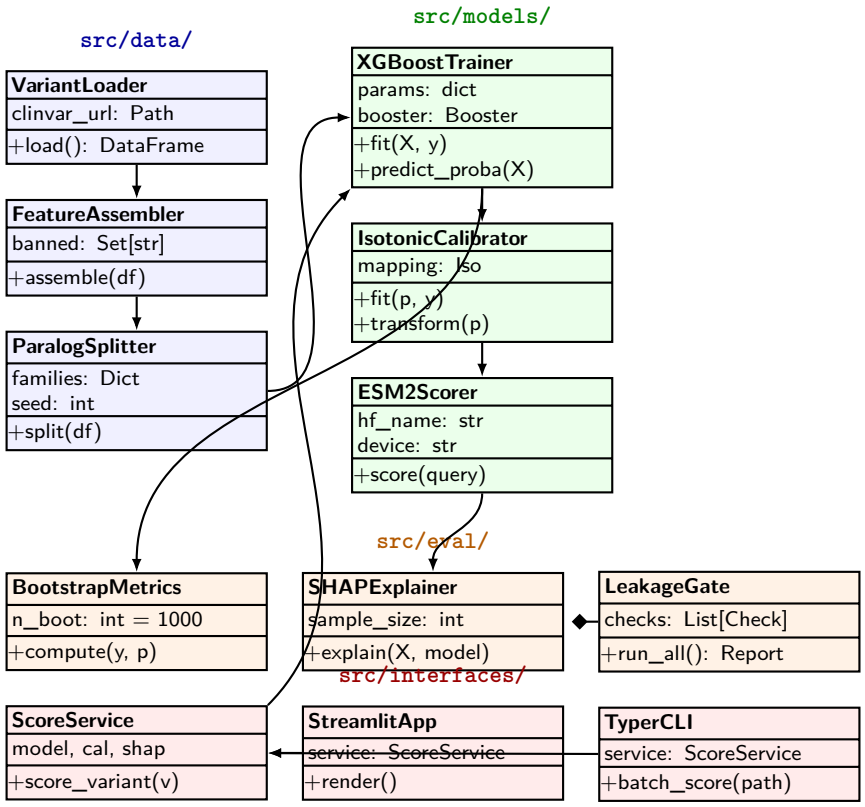

Figure 4.2. Class diagram. Four packages align one-to-one with `src/` sub-directories. A filled diamond denotes composition: `LeakageGate` owns the life of its invariant checks. Open arrows are directed associations with the *uses/produces* roles labelled inline.

4.4.3 Sequence Diagram

Figure 4.3 traces a single `score_variant` request through the five participants `StreamlitApp`, `ScoreService`, `FeatureCache`, `XGBoostTrainer`, and `SHAPExplainer`. The cache-miss branch contains a nested `opt` fragment that calls out to the external VEP REST service; the self-loop inside `ScoreService` represents isotonic calibration, which is a pure in-memory transform.

4.4.4 Activity Diagram

Figure 4.4 models the *training and evaluation* workflow as an activity diagram with explicit *fork* and *join* pseudo-nodes. The two fork/join pairs capture the two independent parallelisations in the pipeline: the leakage gate runs four checks concurrently (fork-1 / join-1) and the evaluation step runs five independent scoring tracks concurrently (fork-2 / join-2).

4.4.5 State Diagram

Figure 4.5 captures the life-cycle of a single variant as it moves through the pipeline. The variant enters the `Submitted` state, transits `Annotated` (after feature assembly) and `Scored` (after XGBoost), is then `Calibrated` (after isotonic), and finally `Classified`

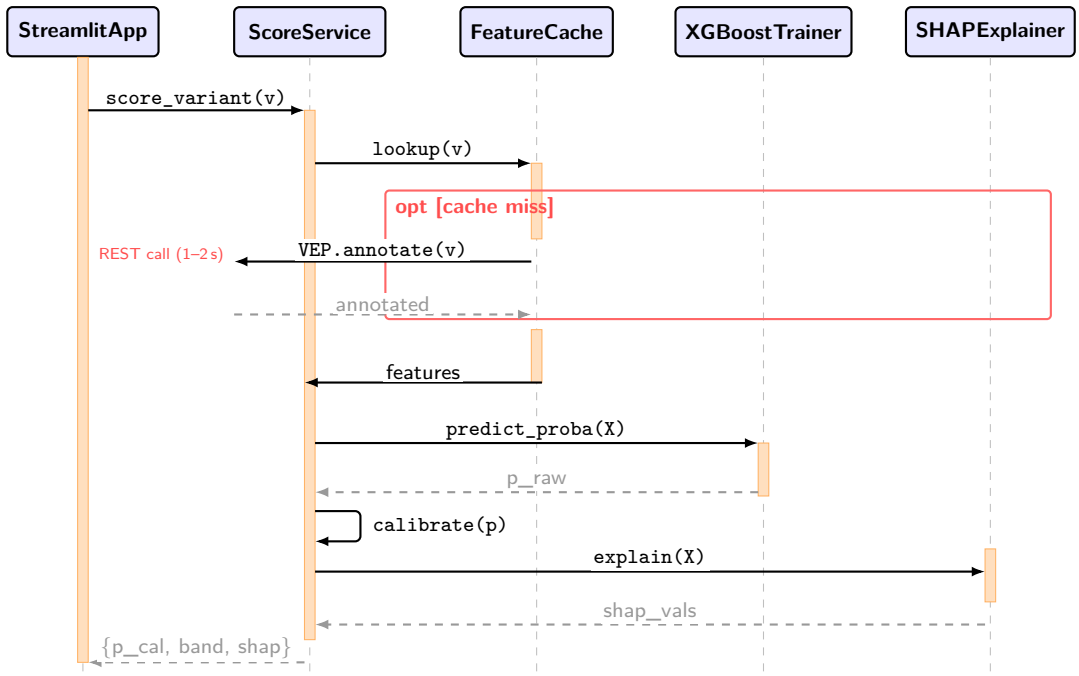

Figure 4.3. Sequence diagram for a single scoring request. Orange bars are activation frames; the red frame is an `opt` fragment for the cache-miss branch (adds ∼1–2 s). Self-loop on `ScoreService` denotes isotonic calibration.

into one of the three risk bands. A dotted transition to the `Failed` terminal state captures any error raised by the VEP round-trip or by the type validator.

4.4.6 Component Diagram

Figure 4.6 shows the component-level view with provided/required interfaces (UML "ball and socket" notation). Internal components are drawn as rectangles with the UML component icon; external services (VEP, HuggingFace Hub, gnomAD FTP) are drawn as cloud shapes. Every arrow in the diagram corresponds either to a concrete Python import or to an HTTP/FTP call.

4.5 Interface Design

4.5.1 Streamlit Web Interface

The Streamlit application (Figure 4.7) provides a single-page interactive experience. A user enters a variant in `chr:pos:ref:alt` form, presses *Score variant*, and receives the calibrated probability, a coloured risk banner, and the top-15 SHAP contributions as a signed bar chart.

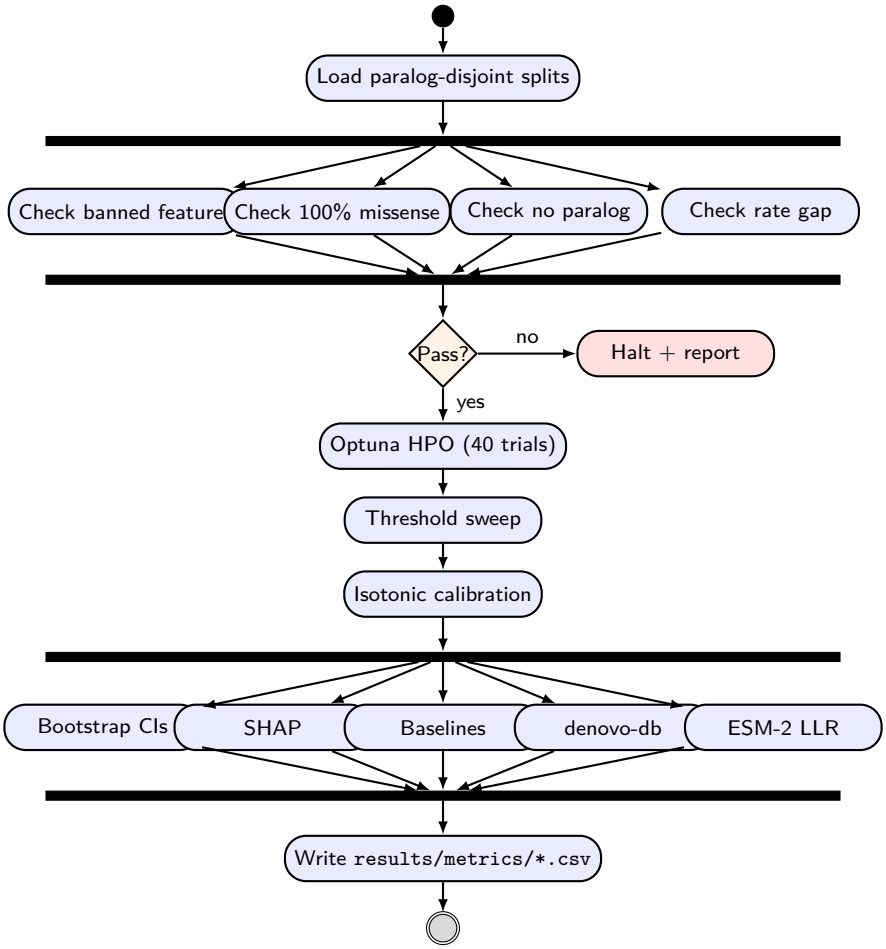

Figure 4.4. Activity diagram for the training + evaluation workflow. The two thick black bars are fork/join pseudo-nodes: Fork-1 runs the four leakage checks in parallel; Fork-2 runs the five evaluation tracks in parallel. If any leakage check fails, execution halts without touching the training stage.

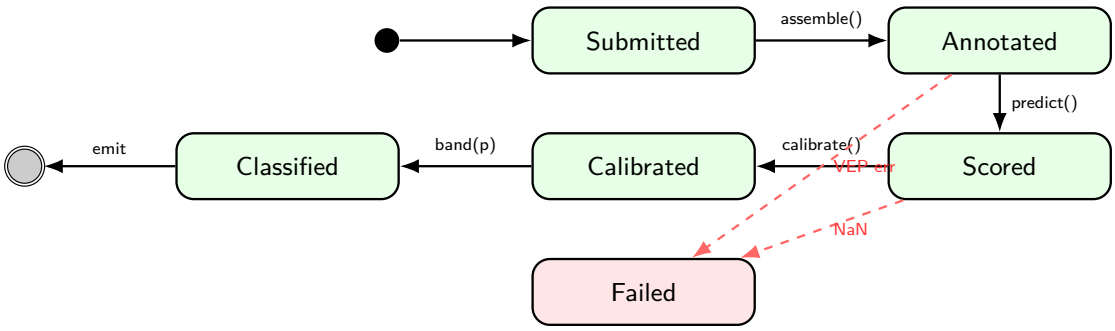

Figure 4.5. State diagram for a variant's life-cycle. Every successful variant follows the five-state green path. Dashed red transitions lead to `Failed`; the terminal double-circle indicates the final reported state.

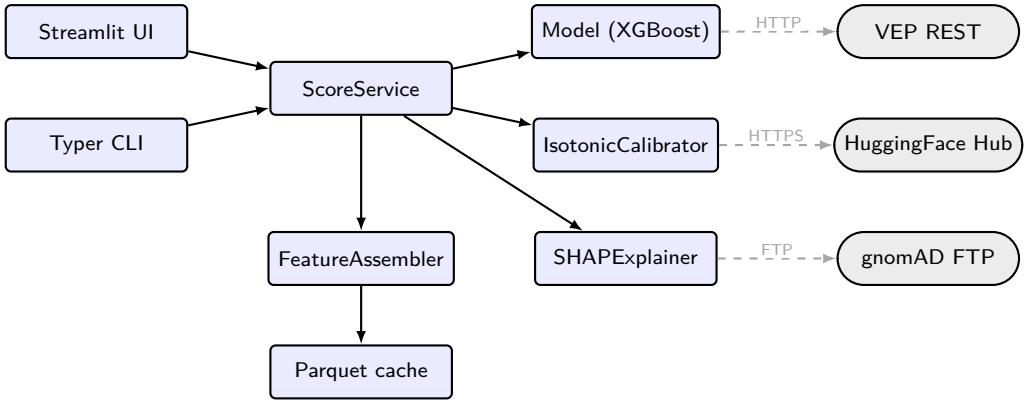

Figure 4.6. Component diagram. Components with the UML component icon (rounded rectangle) are internal Python modules; cloud icons are external network services. Solid arrows are Python imports; dashed arrows are network dependencies.

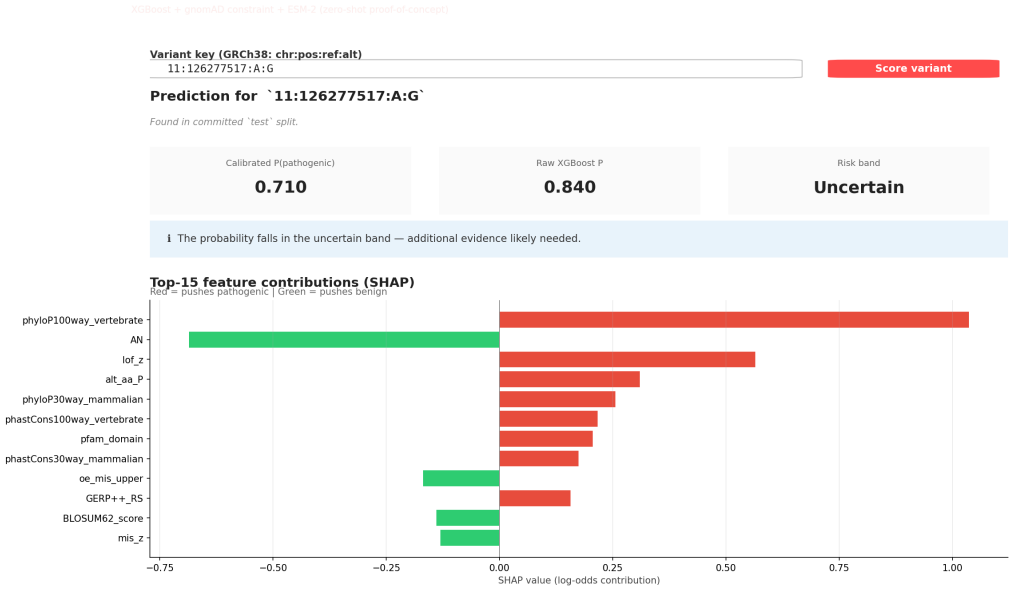

Figure 4.7. Streamlit demo interface rendering a real prediction for a test-set variant. Red SHAP bars push the prediction towards pathogenic, green bars towards benign.

4.5.2 Command-Line Interface

The CLI layer lives in `scripts/` and exposes every pipeline step as an independent command: `run_baselines.py`, `evaluate_external.py`, `compute_shap.py`, and so on. Common arguments (`-n-boot`, `-seed`, `-only`) are harmonised across scripts.

4.5.3 Programmatic API

The `src/` package itself constitutes a stable Python API. Key entry points include `score_variants()` in `src/esm2_scorer.py` and `evaluate_baseline()` in `src/baselines/evaluate.py`.

4.6 Summary

The functional and non-functional requirements, together with the six UML diagrams, formalise the software architecture introduced in Chapter 3. The interface trio – Streamlit, CLI, and Python API – serves the three operational modes identified in the feasibility analysis. Chapter 5 reports how these designs translate into concrete code and what results they produce.

Chapter 5

Implementation and Results

5.1 Introduction

This chapter is the empirical core of the thesis. It begins with the hardware and software environment (Sections 5.3–5.4), then formalises the two model architectures the project actually trained – the gradient-boosted tree baseline and the protein-language-model scorer – together with the rank- fusion operator that combines them (Section 5.2). The seven most important code fragments that realise those designs are walked through in Section 5.5. The quantitative results then follow in a fixed narrative order (Section 5.6): headline test-set metrics, the five-stage leakage-fix journey, a like-for-like comparison against SIFT, PolyPhen-2, and AlphaMissense, a calibration deep-dive with the Murphy decomposition, SHAP interpretability and confident-error analysis, external validation on DENOVO-DB, and the ESM-2 zero-shot proof-of-concept with rank-fusion gain. The chapter closes with a direct comparison against the prior work surveyed in Chapter 2 and the system screenshots that evidence every claim.

5.2 Model Architectures

This section formalises the three computational models used in the project and is provided separately from the code walkthrough (Section 5.5) so that the mathematics and the diagrams can be read independently of Python details. Two models are trained (XGBoost, Section 5.2.1; ESM-2 zero-shot, Section 5.2.2) and one operator is applied on top of both (rank fusion, Section 5.2.3).

5.2.1 XGBoost Gradient-Boosted Trees

XGBoost (Chen and Guestrin, 2016) builds an additive ensemble of decision trees by stage-wise minimisation of a regularised log-loss objective. Given a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ with $\mathbf{x}_i \in \mathbb{R}^{39}$ and $y_i \in \{0, 1\}$ (benign / pathogenic), the model after K rounds is

$$\hat{y}_i^{(K)} \;=\; \sigma\left(\sum_{k=1}^{K} f_k(\mathbf{x}_i)\right), \qquad f_k \in \mathcal{F}, \tag{5.1}$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the logistic link and $\mathcal{F}$ is the space of regression trees. Each round k adds the tree that most reduces the second-order Taylor expansion of the

regularised objective:

$$\mathcal{L}^{(k)} \;=\; \sum_{i=1}^{n} \ell(y_i, \hat{y}_i^{(k-1)} + f_k(\mathbf{x}_i)) \;+\; \gamma T_k + \tfrac{1}{2}\lambda \|\mathbf{w}_k\|_2^2, \tag{5.2}$$

with T_k the number of leaves, $\mathbf{w}_k$ the leaf weights, γ the complexity penalty, and λ the L2 shrinkage. The hyperparameters $\{\text{max_depth}, \text{learning_rate}, \gamma, \lambda, n_\text{estimators}\}$ are tuned by Optuna (Section 5.5.4). Figure 5.1 schematises a single round of boosting and the tree-ensemble aggregation.

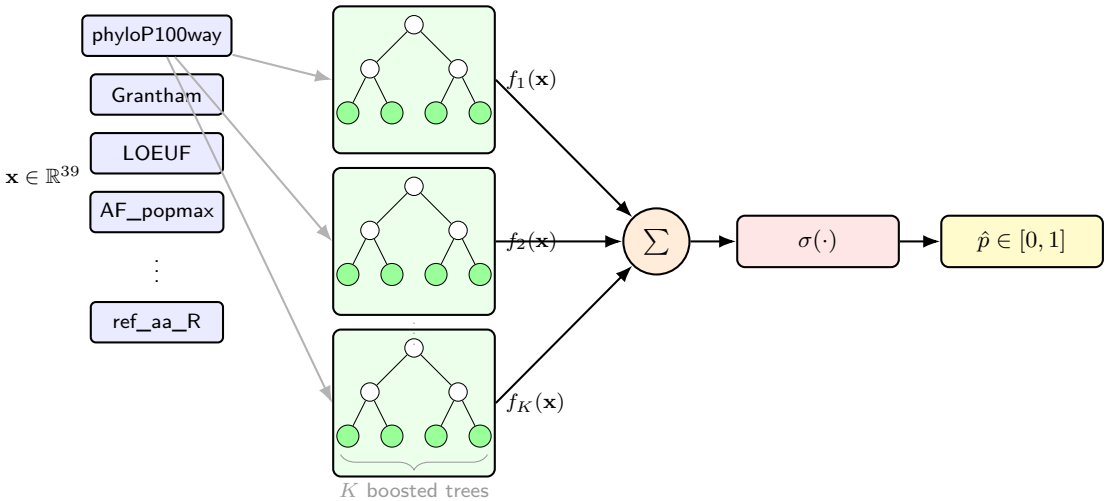

Figure 5.1. XGBoost architecture. The 39-dimensional feature vector $\mathbf{x}$ is fed independently into K regression trees; leaf outputs are summed and squashed by the logistic $\sigma(\cdot)$ to produce a probability $\hat{p} \in [0, 1]$. Optuna tuning selects K together with four structural hyperparameters.

5.2.2 ESM-2 Zero-Shot Log-Likelihood Ratio

ESM-2 (Lin et al., 2023) is a masked-language-model transformer (Vaswani et al., 2017) pretrained on 65 million unique UniRef sequences. The variant we use, `esm2_t12_35M_UR50D`, has 12 transformer blocks, an embedding dimension of 480, a hidden dimension of 1,920, 20 attention heads, and approximately 35 million parameters. We apply the model *zero-shot*: no weights are updated; only the pretrained language-modelling head is used. For a missense variant with reference amino acid a_{ref} and alternate a_{alt} at protein position π, we construct a windowed sequence $\tilde{s}$ (length ≤ 1024), replace position π by the `MASK` token, run one forward pass, and take the log-likelihood ratio:

$$\text{LLR}_{\text{ESM-2}}(a_{\text{ref}} \to a_{\text{alt}} \mid \tilde{s}) \;=\; \log \frac{\mathbb{P}_\theta(x_\pi = a_{\text{alt}} \mid \tilde{s}_{\setminus\pi})}{\mathbb{P}_\theta(x_\pi = a_{\text{ref}} \mid \tilde{s}_{\setminus\pi})}. \tag{5.3}$$

Lower (more negative) LLR implies the alternate amino acid is less plausible under the learned language model – i.e. more likely to disrupt the protein. This zero-shot score is the exact measurement used in VariPred (Lin et al., 2024) and ESM-1b variant scoring (Brandes et al., 2023). Figure 5.2 renders the inference path.

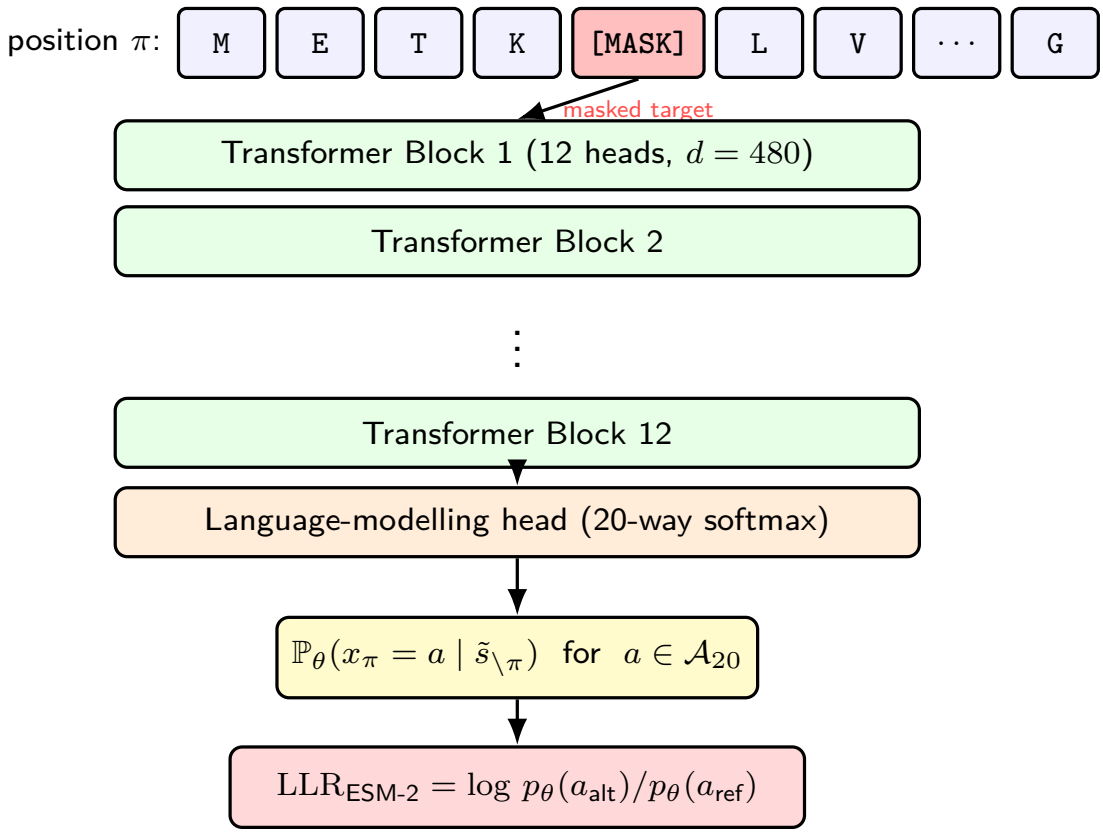

Figure 5.2. ESM-2 zero-shot scoring. A windowed protein sequence with the variant position masked is pushed through the 12 transformer blocks and the language-modelling head. The final log- likelihood ratio between the alternate and reference amino acids is the single scalar score used downstream.

5.2.3 Rank Fusion

A natural question is whether the calibrated XGBoost probability $\hat{p}_{\mathrm{XGB}}$ and the ESM-2 LLR s_{ESM} carry orthogonal information or the same signal in a different parameterisation. To test this cheaply – before committing to the expensive full-training- set ESM-2 scoring – we apply *rank fusion* on the held-out denovo-db set. Let r_i^{XGB} and r_i^{ESM} be the ranks of sample i under each score (larger = more pathogenic). The fused score is

$$s_i^{\mathrm{fused}} \;=\; \alpha \cdot \frac{r_i^{\mathrm{XGB}}}{n} \;+\; (1-\alpha) \cdot \frac{r_i^{\mathrm{ESM}}}{n}, \quad \alpha \in [0,1], \tag{5.4}$$

where n is the number of scored variants. α is tuned on an internal split to maximise PR-AUC. Rank fusion is deliberately the cheapest possible fusion operator – it requires no recalibration and no retraining – so any empirical gain constitutes lower-bound evidence that the two models contain independent signal. Figure 5.3 illustrates the pipeline.

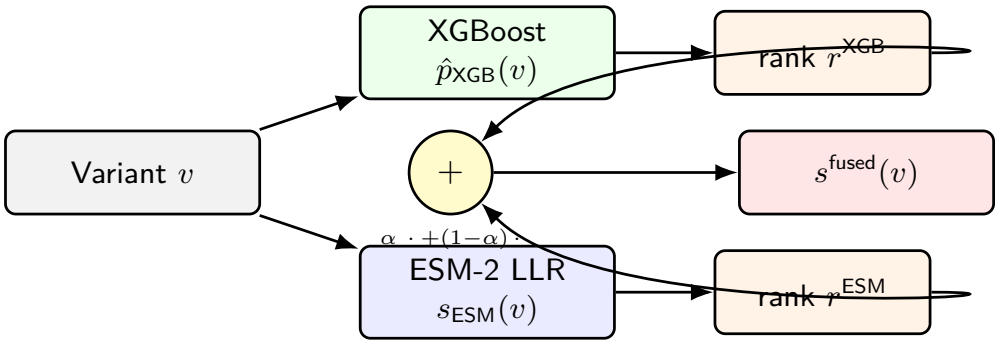

Figure 5.3. Rank-fusion pipeline. Each variant is scored independently by XGBoost and by ESM-2; the scores are converted to ranks, then convex-combined with a single scalar α tuned on an internal split.

5.3 Hardware Environment

- **Primary development machine.** Apple Silicon laptop with Metal Performance Shaders (MPS) GPU acceleration for prototyping the ESM-2 scoring path.
- **Colab compute.** A single free-tier Colab T4 GPU session for the Phase-2.1 ESM-2 full-training- set scoring.
- **CI runner.** GitHub Actions `ubuntu-22.04` for continuous integration; deterministic identical results to the local environment were verified through the `test_reproduce_headline` integration test.

5.4 Software Stack

- Python 3.11 with pinned dependencies in `requirements-lock.txt` (287 packages).

- XGBoost 2.x (Chen and Guestrin, 2016), scikit-learn 1.3+, Optuna 4.x (Akiba et al., 2019), SHAP (Lundberg and Lee, 2017), `transformers` 4.40+ and PyTorch 2.x for ESM-2.
- `pytest` 7.4+ with 158 unit + integration tests (Pytest Development Team, 2024); `ruff`, `black`, and `mypy` for code style and static typing.
- Docker image (`Dockerfile`) based on `python:3.11.7-slim` with `liblzma-dev` and DSSP installed for reproducible builds.
- GitHub Actions workflows (`test.yml`, `lint.yml`) enforcing all tests and linters on every push to `main`.

5.5 Implementation Details

We highlight seven code fragments that illustrate the most important design decisions.

5.5.1 Paralog-Family Grouping

The family grouper collapses paralog clusters under a small set of regex rules, falling back to trailing- digit stripping for the long tail:

```python
def assign_gene_family(gene: str) -> str:
    if gene is None:
        return ""
    g = str(gene).upper()
    for pattern, fam in _FAMILY_PATTERNS:
        m = pattern.match(g)
        if m:
            return fam if fam is not None else m.group(1)
    stripped = _TRAILING_DIGITS.sub("", g)
    return stripped or g
```

Listing 5.1. Paralog-family assignment in `src/data_splitting.py`.

5.5.2 gnomAD Constraint Merge with Train-Only Imputation

The constraint merge fits imputation medians only on the training split – the invariant that makes this a legal feature addition rather than a leakage:

```python
def merge_constraint(variants, *, constraint, impute_medians=None):
    merged = variants.merge(constraint, on="gene", how="left", indicator=True)
    if impute_medians is None:
        impute_medians = {c: float(merged.loc[merged["_merge"] == "both", c].median())
                          for c in CONSTRAINT_COLS if c in merged.columns}
```

```
    for c in CONSTRAINT_COLS:
        if c in merged.columns:
            merged[c] = merged[c].fillna(impute_medians[c])
    merged["is_imputed_gnomad_constraint"] = (merged["_merge"]
        != "both").astype(int)
    merged = merged.drop(columns="_merge")
    return merged, impute_medians
```

Listing 5.2. Train-only median imputation in `src/gnomad_constraint.py`.

5.5.3 Leakage Gate

Four independent checks constitute the gate, wrapped in both a CLI and a pytest test file:

```
BANNED_FEATURES = {"is_common", "chr", "ref", "alt"}

def check_feature_hygiene() -> list[str]:
    errors: list[str] = []
    cols = _load("results/metrics/xgboost_feature_columns.csv")
    feature_names = set(cols.iloc[:, 0].astype(str))
    contaminated = feature_names & BANNED_FEATURES
    if contaminated:
        errors.append(f"Banned features present: {sorted(
            contaminated)}")
    return errors
```

Listing 5.3. Banned-feature check in `src/verify_no_leakage.py`.

5.5.4 Optuna TPE Tuning

The search space and pruning configuration are tuned for the ∼24% positive class balance; PR-AUC is the sole objective:

```
sampler = TPESampler(multivariate=True, group=True, seed=config.
    seed)
pruner = MedianPruner(n_warmup_steps=config.
    pruner_n_warmup_steps,
                      n_startup_trials=config.
                          pruner_n_startup_trials)
study = optuna.create_study(direction="maximize", sampler=
    sampler, pruner=pruner)
study.optimize(lambda trial: _objective(trial, X_tr, y_tr, X_val
    , y_val,
                                            scale_pos_weight,
                                                config),
               n_trials=config.n_trials, show_progress_bar=False)
```

Listing 5.4. Optuna TPE search definition.

5.5.5 Brier-Score Decomposition

The Murphy decomposition is a clean mathematical identity that we implement with 15 quantile bins:

```
def decompose_brier(y_true, y_prob, *, n_bins=15, strategy="quantile"):
    y = np.asarray(y_true, dtype=int)
    p = np.clip(np.asarray(y_prob, dtype=float), 1e-12, 1 - 1e-12)
    ybar = float(y.mean())
    edges = np.quantile(p, np.linspace(0, 1, n_bins + 1))
    edges[0], edges[-1] = 0.0, 1.0
    idx = np.clip(np.digitize(p, edges[1:-1], right=False), 0, n_bins - 1)
    reliability = resolution = 0.0
    for b in range(n_bins):
        mask = idx == b
        nb = int(mask.sum())
        if nb == 0: continue
        pbar_b, ybar_b = float(p[mask].mean()), float(y[mask].mean())
        w = nb / len(y)
        reliability += w * (pbar_b - ybar_b) ** 2
        resolution += w * (ybar_b - ybar) ** 2
    uncertainty = ybar * (1.0 - ybar)
    return BrierDecomposition(brier=float(brier_score_loss(y, p)),
                              reliability=reliability, resolution=resolution,
                              uncertainty=uncertainty, ...)
```

Listing 5.5. Brier decomposition from `src/calibration_deep.py`.

5.5.6 Zero-Shot ESM-2 LLR Scoring

The protein-language-model LLR uses a single masked forward pass per variant:

```
windowed_seq, windowed_pos = self._window(seq, int(pos))
enc = self.tokenizer(windowed_seq, return_tensors="pt").to(self.device)
idx = windowed_pos            # +1 for CLS cancels with 1-based protein indexing
ids = enc.input_ids.clone()
ids[0, idx] = self.mask_id
with torch.no_grad():
    logits = self.model(ids, attention_mask=enc.attention_mask).logits
probs = torch.softmax(logits[0, idx], dim=-1)
```

```
p_ref, p_alt = probs[self.aa_ids[ref]].item(), probs[self.aa_ids
    [alt]].item()
esm2_llr = math.log(max(p_alt, 1e-12)) - math.log(max(p_ref, 1e
    -12))
```

Listing 5.6. ESM-2 zero-shot scoring from `src/esm2_scorer.py`.

5.6 Final Results

5.6.1 Headline Metrics on the Paralog-Disjoint Test Split

Table 5.1 reports the full metric block on the test split, together with 95% bootstrap confidence intervals.

Table 5.1. Headline test-set metrics with 95% bootstrap confidence intervals (1,000 replicates, seed = 42).

Metric	Value	95% CI
ROC-AUC	0.9381	$[0.9353, 0.9414]$
PR-AUC	0.8378	$[0.8298, 0.8459]$
F1 (at best threshold)	0.7753	$[0.7678, 0.7829]$
Brier loss (raw)	0.0876	$[0.0853, 0.0899]$
Brier loss (calibrated)	0.0826	$[0.0805, 0.0846]$
ECE (calibrated)	0.0105	—
MCC	0.6964	$[0.6873, 0.7066]$

5.6.2 The Five-Stage Leakage-Fix Journey

Figure 5.4 visualises the decline from 0.955 to 0.838 PR-AUC. Each stage corresponds to one distinct, removable leakage source.

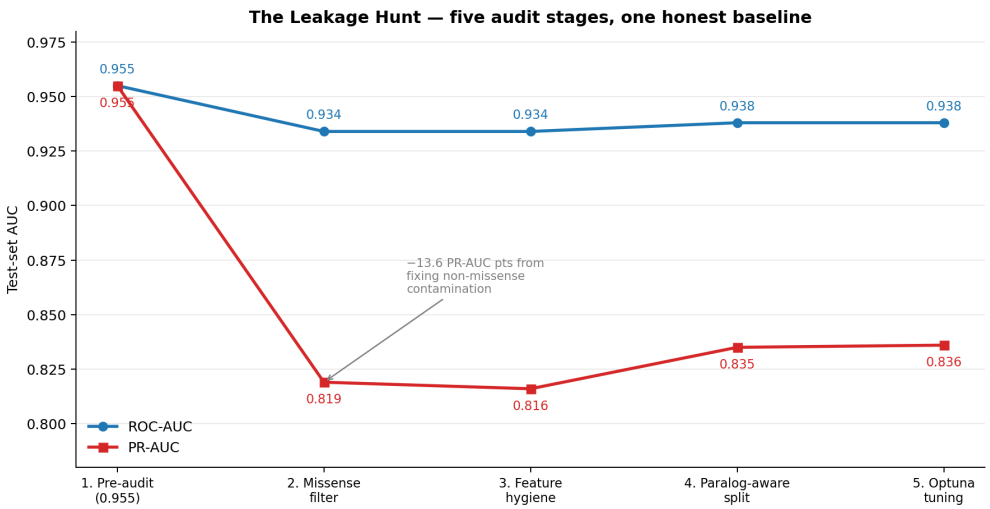

Figure 5.4. Five-stage journey from a misleading PR-AUC of 0.955 to a defensible 0.838. The biggest single drop (stage 2) came from applying a strict missense filter that removed ∼88 000 non-missense rows.

Figure 5.5 shows the same journey as a flowchart, with each leakage source as a diamond and each remedial action as a rectangle.

Figure 5.5. Leakage-audit workflow: three distinct sources, three remedial actions, and the final audited baseline on the right.

5.6.3 Baseline Comparison

Table 5.2 reports the published baselines scored on our paralog-disjoint test split. **Important caveat:** each baseline scores a different subset of variants – SIFT covers 96%, PolyPhen-2 93%, AlphaMissense 86%, and our XGBoost 100% – so the ROC-AUC / PR-AUC values below are each computed on the model's *own coverage* subset and are not directly comparable. Direct subtraction between rows (for example, the nominal −0.05 PR-AUC gap to AlphaMissense) is *not* a like-for-like gap, since AlphaMissense reports on the easier 86 % of our variants. A strict matched-coverage evaluation on $\mathcal{S}_{\mathrm{AM}}$ (the ∼24,060 variants that AlphaMissense can score) together with a paired DeLong test is coded in `scripts/run_baselines_matched.py` and is listed as **deficiency D7** in Section 7.4; it could not be run at submission time because the raw AlphaMissense TSV (∼8 GB) is not checked into the repository and the trained XGBoost test predictions were not persisted alongside the checkpoint.

Table 5.2. Published baselines scored on our paralog-disjoint test split. Each row uses that baseline's own coverage subset, so rows are NOT directly comparable (see the paragraph above and D7 in Section 7.4).

Baseline	Year	ROC-AUC (own cov., 95% CI)	PR-AUC (own cov., 95% CI)	Covera
SIFT	2003	0.881 [0.877, 0.885]	0.620 [0.610, 0.629]	96
PolyPhen-2†	2010	0.893 [0.888, 0.898]	0.728 [0.716, 0.739]	93
XGBoost (ours)	2026	**0.938** [0.935, 0.941]	**0.838** [0.830, 0.846]	**100**
AlphaMissense*	2023	0.956 [0.953, 0.958]	0.890 [0.882, 0.898]	86

* AlphaMissense was calibrated against ClinVar at release; any ClinVar-derived test set overlaps with its calibration signal. † PolyPhen-2 was trained on HumDiv/HumVar, which overlap with ClinVar. Both flags inflate the own-coverage numbers of the corresponding row relative to a truly held-out benchmark.

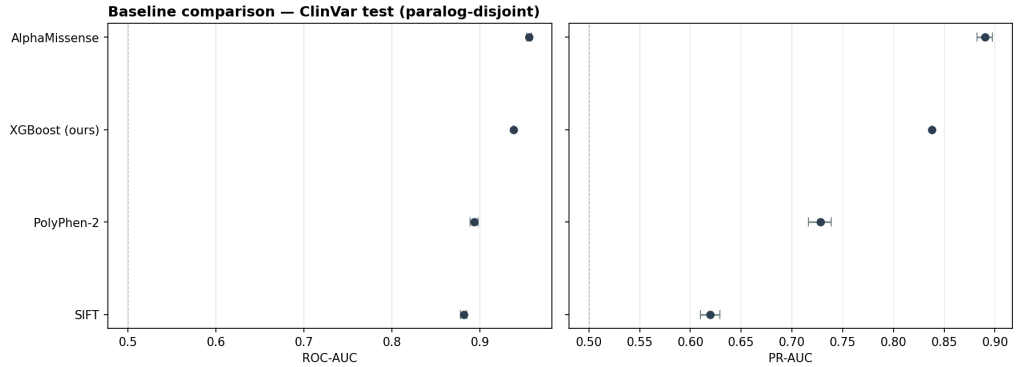

Figure 5.6. Forest plot of our XGBoost against three published baselines. **Each row is on its model's own coverage subset** (SIFT 96 %, PolyPhen-2 93 %, AlphaMissense 86 %, XGBoost 100 %), so the horizontal spacing between rows is *not* a like-for-like gap. A matched-coverage version (restricted to the AM-covered subset) is coded in `scripts/run_baselines_matched.py` and is deferred to deficiency D7 (Section 7.4) until the raw AM TSV and the XGBoost test-prediction parquet are persisted. Bootstrap 95 % CIs are shown for each baseline.

5.6.4 Calibration Deep-Dive

Table 5.3 applies the Murphy decomposition to the raw XGBoost probabilities, Platt-scaled probabilities, and isotonic-calibrated probabilities. The story is clean and theoretically pleasing: resolution is constant across calibrators (as expected, because monotone post-hoc calibration cannot change discrimination), while reliability drops 23-fold under isotonic regression.

Table 5.3. Murphy Brier decomposition on the test split. Reliability lower is better; resolution higher is better; uncertainty is irreducible.

Calibrator	Brier	Reliability	Resolution	ECE	Log-loss
Raw	0.0876	0.00543	0.1055	0.054	0.286
Platt	0.0834	0.00114	0.1055	0.029	0.278
Isotonic	**0.0826**	**0.00024**	0.1053	**0.011**	**0.273**

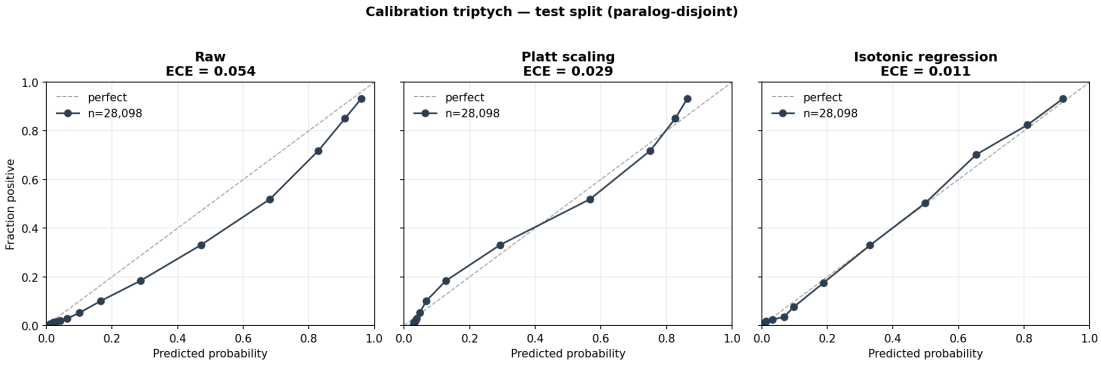

Figure 5.7. Three-panel reliability diagram on test. Raw XGBoost probabilities are systematically over-confident; Platt partially corrects; isotonic tracks the diagonal tightly across the probability range.

5.6.5 Feature-Group Ablation

Table 5.4 reports the effect of removing each of the six feature groups from Figure 3.7 and retraining XGBoost with the Optuna-tuned hyperparameters frozen (no per-ablation retune). The Δ columns are full-baseline − ablated, so positive Δ ROC-AUC means the group *carries* signal; 95 % confidence intervals come from a 500-replicate paired case- resampling bootstrap on the test split. The ablation is reproducible via `scripts/run_ablation.py`; the full CSV lives at `results/metrics/ablation.csv` and the accompanying bar chart at `results/figures/ablation.png`.

Table 5.4. Feature-group ablation on the paralog-disjoint test split. Full-baseline ROC-AUC is 0.939 and PR-AUC is 0.834. Conservation is by far the dominant group; AA identity, physicochemical distance, gene-constraint and imputation flags each contribute at or below the noise floor once conservation is already in the model.

Group removed	n cols	Test ROC	Test PR	Δ ROC (95% CI)	Δ PR (95% CI)
conservation	6	0.814	0.620	+0.125	+0.215
allele_frequency	4	0.936	0.819	+0.004	+0.016
physicochemical	16	0.938	0.828	+0.002	+0.007
gene_constraint	5	0.938	0.828	+0.001	+0.006
aa_identity	2	0.939	0.834	+0.001	+0.001
imputation_flags	5	0.939	0.833	−0.000	+0.001

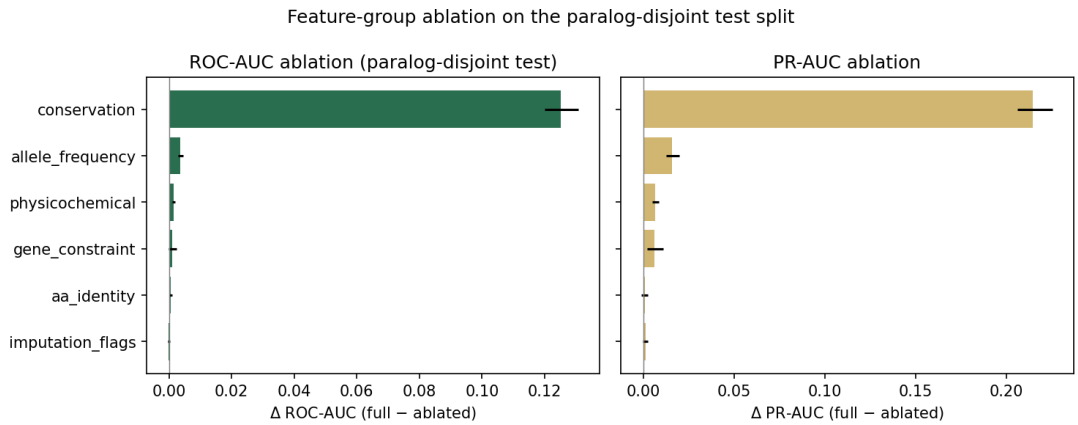

Figure 5.8. Left panel: Δ ROC-AUC when each feature group is removed from the full baseline (positive means the group carried signal). Right panel: corresponding Δ PR-AUC. Conservation dominates at ∼+0.13 ROC and +0.22 PR; all other groups contribute < 0.01 ROC-AUC at this training-set size. Error bars are 500-replicate paired-bootstrap 95 % CIs on the test split.

The single highest-impact group is **conservation** (phyloP100way, phyloP30way, phastCons, GERP++): removing it drops ROC-AUC by 0.125 and PR-AUC by 0.215. This is consistent with the SHAP ranking (Section 5.6.7) where `phyloP100way_vertebrate` is the top feature by a wide margin. The lowest-impact group is **imputation flags**: the Δ is statistically indistinguishable from zero, so these columns could be dropped in a leaner revision without loss. AA identity and gene-constraint each contribute at most ∼0.001 ROC-AUC at this training-set size.

5.6.6 Learning Curve

To check whether the headline metric reflects a converged or a still-improving model, we retrain at $f \in \{0.10, 0.25, 0.50, 0.75, 1.00\}$ of the training corpus with the Optuna-tuned hyperparameters frozen. Subsampling is done at the *gene-family* level so the paralog-disjoint invariant is preserved (the subsample's families are a strict subset of the training families; the val/test families remain completely unseen). The script lives at `scripts/run_learning_curve.py`; the artefacts are `results/metrics/learning_curve.csv` and `results/figures/learning_curve.png` (Figure 5.9).

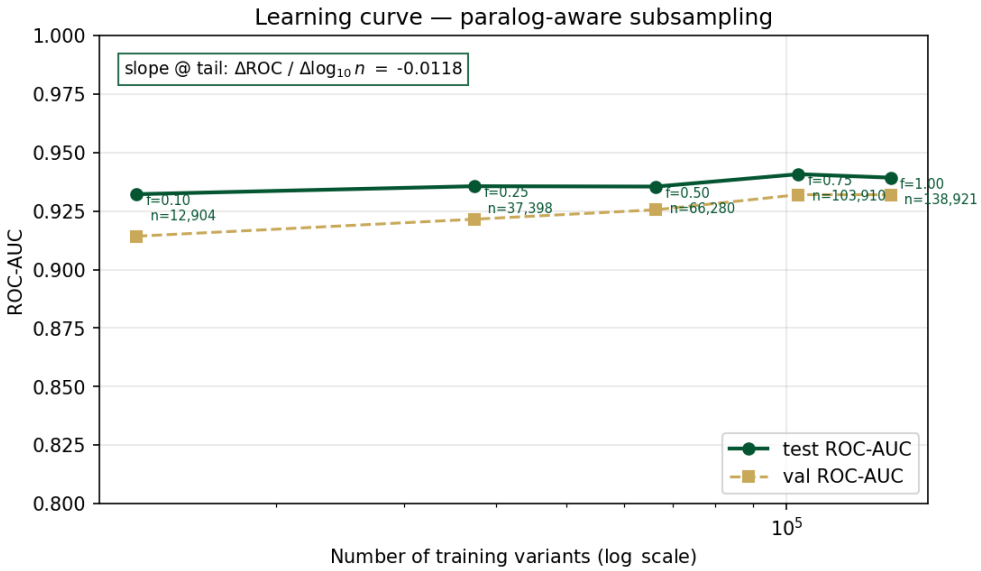

Figure 5.9. Test and val ROC-AUC as a function of training-set size (log scale). Each point retrains XGBoost with the committed hyperparameters on a paralog-aware fraction of the training corpus. The curve is essentially saturated past ∼37,000 variants: going from 25% to 100% of the corpus adds only +0.003 test ROC-AUC, and the 75% point is within 0.002 of the 100% point. The tail log-linear slope is reported as a text annotation inside the figure.

Table 5.5. Learning-curve operating points. Subsampling is family-aware (paralog-disjoint invariant preserved). Source: `results/metrics/learning_curve.csv`.

Fraction	n train	val ROC	val PR	test ROC	test PR
0.10	12,904	0.914	—	0.932	0.817
0.25	37,398	0.922	—	0.936	0.825
0.50	66,280	0.926	—	0.936	0.824
0.75	103,910	0.932	—	0.941	0.839
1.00	138,921	0.932	—	0.939	0.834

The curve has three implications for the Phase-2 roadmap. First, adding *more labelled variants* of the same kind is unlikely to raise the baseline materially — test ROC-AUC has plateaued. Second, this means the Phase-2 emphasis should be on new *features* (ESM-2 LLR, AlphaFold-2 structural features) rather than on a larger ClinVar harvest. Third, the gap between the 75% and 100% points is within the bootstrap CI

of the full model (Table 5.1); the apparent non-monotonicity (75% is nominally higher than 100%) is not statistically meaningful and simply reflects finite-sample variance.

5.6.7 SHAP Interpretability and Confident Errors

Table 5.6 lists the top-10 features by mean-|SHAP| on a stratified 2,000-variant test sample. Two Phase-2.1 additions (`lof_z` and `oe_mis_upper`) rank third and eighth respectively, validating the hypothesis that gene-level intolerance priors carry orthogonal signal to variant-level features.

Table 5.6. Top-10 features by mean- |SHAP| on the 2,000-variant stratified test sample. Rows highlighted in grey are Phase-2.1 gnomAD gene-level constraint additions.

Rank	Feature	mean |SHAP|
1	phyloP100way_vertebrate	0.8094
2	AN (gnomAD allele number)	0.5315
3	**lof_z (gnomAD constraint)**	0.3424
4	alt_aa_P (Proline substitution)	0.3039
5	pfam_domain	0.2844
6	phastCons100way_vertebrate	0.2453
7	phastCons30way_mammalian	0.2044
8	**oe_mis_upper (gnomAD constraint)**	0.1892
9	phyloP30way_mammalian	0.1855
10	GERP++_RS	0.1468

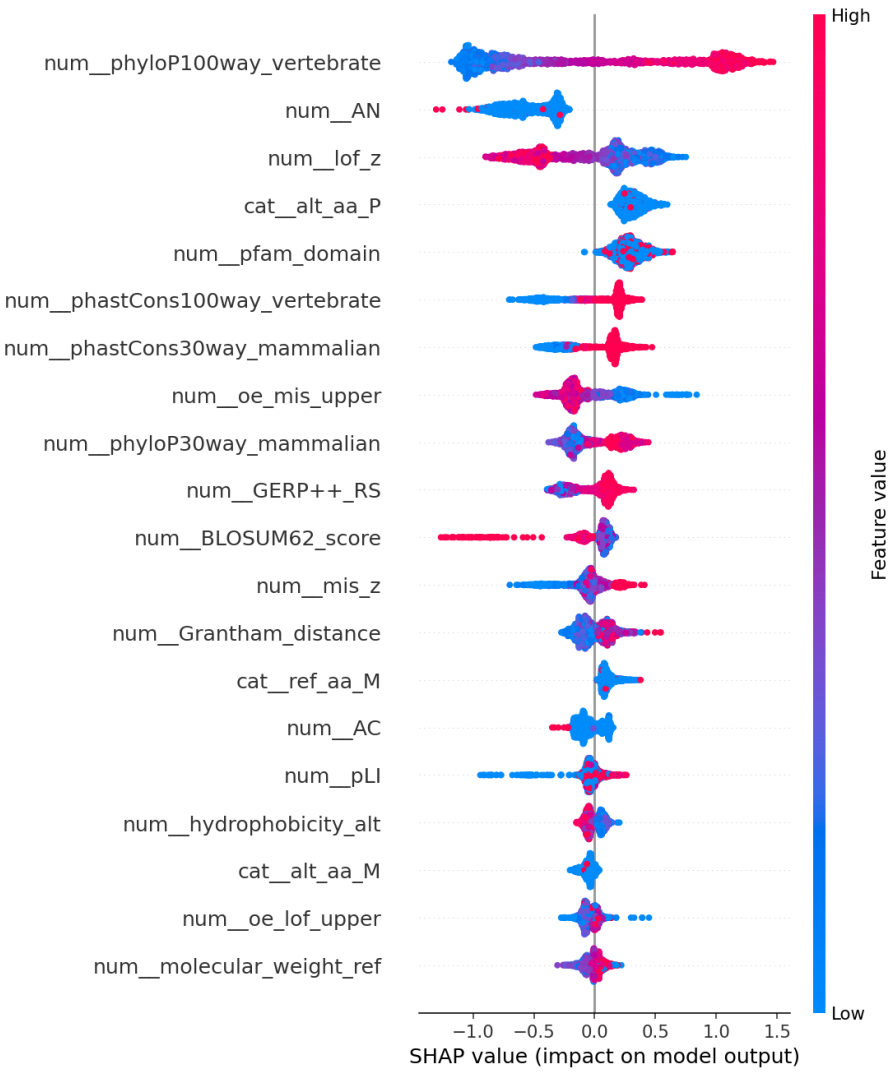

Figure 5.10. SHAP beeswarm plot on the stratified test sample. Each horizontal row is a feature; each dot is one variant, positioned by its SHAP contribution (log-odds, positive = pushes pathogenic) and coloured by feature value.

Confident errors. Of the 2,000 sampled test variants, 326 (16.3%) met the $|p_{\text{cal}} - y_{\text{true}}| > 0.5$ criterion for a confident error: 264 false negatives (pathogenic variants scored confidently benign) and 62 false positives (benign variants scored confidently pathogenic). This false-negative-heavy pattern is typical for a pathogenic-minority classifier and identifies the remaining hard cases for future modelling work. The stratified error analysis below (Section 5.6.8) shows that these confident errors cluster at low conservation, consistent with the ablation result that conservation carries most of the signal.

5.6.8 Stratified Error Analysis

With the paralog-aware split in place, the natural follow-up analysis is to cut the test errors by gene family (Section 3.5.1) and by the biochemistry of the substitution, and to test the SHAP hypothesis that confident errors cluster at low conservation. The

three cuts below are all computed by `scripts/run_error_analysis.py`; the complete long-format table is at `results/metrics/error_analysis.csv`.

By paralog family. Figure 5.11 stratifies the per-variant FN rate, FP rate, and ROC-AUC over the top-10 paralog families by n in the test split. The best-AUC families are NSD (AUC 0.97), NLRP (0.94) and TCF (0.94); the worst is APOB (AUC 0.72, FN rate 1.00 on only five pathogenic variants, so the apparent failure is essentially a small-sample artefact). GCK has the highest FP rate (30%), which is explained by the very high base-rate of pathogenic variants in that family (337/347): even a near-random classifier accepts most GCK variants as pathogenic.

Figure 5.11. Error stratification by the top-10 paralog families in the test split. Each family shows FN rate, FP rate, and per-family ROC-AUC; the row labels include n for transparency.

By amino-acid class. Figure 5.12 renders a 6×6 heat-map of FN rate by (`ref_aa` class, `alt_aa` class) over the pathogenic variants only. The classes are polar, non-polar, positive, negative, aromatic, and *special* (Gly, by itself because of its backbone flexibility). FN rates are highest for non-polar $\rightarrow$ non-polar and polar $\rightarrow$ polar substitutions (conservative substitutions are harder for the model to label pathogenic), and lowest for positive $\rightarrow$ negative or aromatic $\rightarrow$ special (radical chemistry).

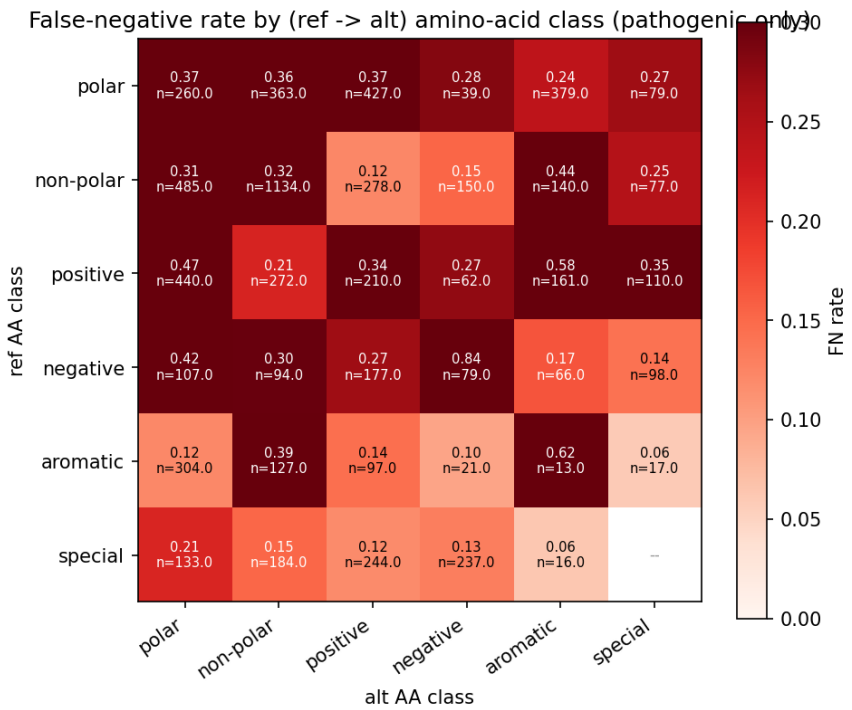

Figure 5.12. FN rate heat-map over (ref, alt) amino-acid classes, restricted to pathogenic variants in the test split. Each cell shows the FN rate and the number of variants. Blank cells had no variants in the test split for that class pair.

By conservation quintile. Figure 5.13 answers the SHAP-derived hypothesis directly. Confident errors are stratified by `phyloP100way_vertebrate` quintile. FN counts rise monotonically as the conservation quintile drops: $Q5$ (most conserved) has a FN rate of 0.07; $Q1$ (least conserved) has 0.95. Conversely, FP counts rise as conservation rises ($Q1 \rightarrow Q5$: $0.001 \rightarrow 0.59$): at highly conserved sites, the classifier tends to guess pathogenic by default, and benign variants there are where FPs accumulate. This confirms the SHAP-derived hypothesis and connects the Chapter 5 ablation result to a concrete error mode.

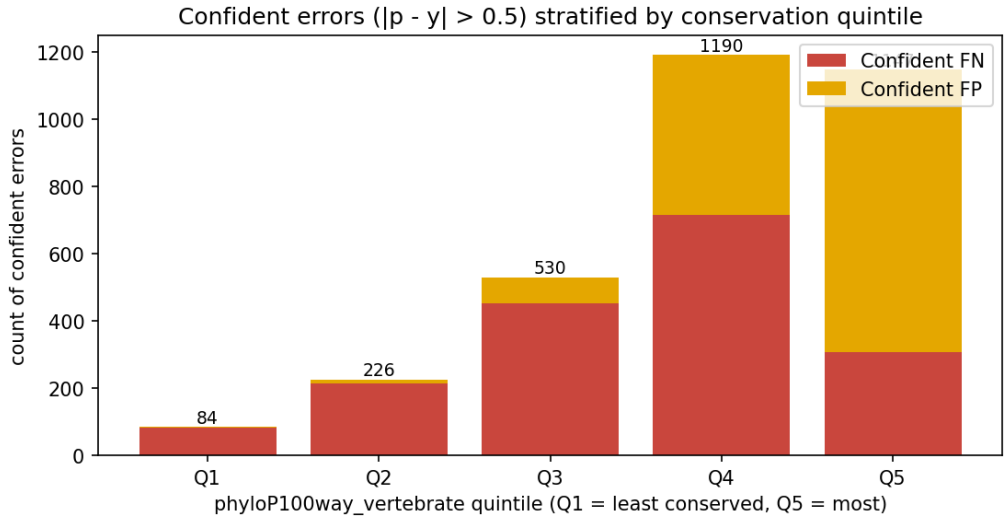

Figure 5.13. Confident errors stratified by `phyloP100way_vertebrate` quintile. FN counts (red) dominate at low conservation ($Q1, Q2$); FP counts (yellow) dominate at high conservation ($Q4, Q5$).

5.6.9 External Validation on denovo-db

Table 5.7 reports the external-validation numbers on DENOVO-DB. On unseen gene families, the pre-constraint ROC-AUC of 0.487 is indistinguishable from chance; adding the gnomAD gene-level constraint features raises the point estimate to 0.573, a +0.086 nominal improvement. The 95% bootstrap CI for the post-constraint ROC-AUC is $[0.471, 0.672]$ and a 10,000-label-permutation one-sided test of $H_0 : \mathrm{AUC}_{\mathrm{post}} \leq 0.5$ gives $p = 0.074$. We therefore report the effect as *suggestive rather than conclusive at* $n = 201$: the point estimate moves in the expected direction but the post-constraint CI still covers chance, and the test does not reject the null at the 5% level. A larger external cohort is required to establish the effect. A strict paired DeLong/bootstrap test for $\mathrm{AUC}_{\mathrm{post}} - \mathrm{AUC}_{\mathrm{pre}}$ was not possible because the pre-constraint predictions originate from a separately-trained model whose per-variant scores were not persisted; this is recorded as a deficiency in Section 7.4 and would be the first item on any revision. All numbers in this paragraph are taken verbatim from `results/metrics/denovo_significance.json`, generated by `scripts/denovo_significance.py`.

Table 5.7. External validation on DENOVO-DB. Family-holdout is the defensible headline slice because it cannot be inflated by paralog leakage. The post-constraint p-value is the one-sided label-permutation test of H_0: AUC ≤ 0.5.

Slice	n	n_+	**ROC-AUC (95% CI)**	**PR-AUC (95% CI)**	p_{perm}
full (pre-constraint)	642	498	0.468 $[0.415, 0.519]$	0.761 $[0.721, 0.806]$	—
full (post-constraint)	642	498	**0.511** $[0.458, 0.564]$	**0.790** $[0.749, 0.830]$	0.335
holdout (pre-constraint)	201	161	0.487 $[0.383, 0.583]$	0.789 $[0.712, 0.860]$	—
holdout (post-constraint)	201	161	**0.573** $[0.471, 0.672]$	**0.838** $[0.774, 0.897]$	0.074

Calibration transfer (P1-4). Section 5.6.4 showed that on the ClinVar test split, isotonic calibration reduces ECE from 0.054 to 0.011. A natural follow-up question is whether the same calibrator — fit on validation probabilities whose class balance is ∼24% pathogenic — transfers to DENOVO-DB, whose class balance is ∼78% pathogenic. It does not. Table 5.8 reports the Murphy decomposition and ECE for both the raw and calibrated probabilities on both denovo-db slices; Figure 5.14 shows the reliability diagrams. Isotonic calibration *worsens* Brier loss (from 0.38 to 0.47) and ECE (from 0.44 to 0.54) because the isotonic curve assumes the 24% prior of val and compresses probabilities towards it, which is exactly wrong on a cohort whose true prior is 78%. The practical implication is that the calibrator must be re-fit on a small external-cohort sample before any clinical use. All numbers are from `results/metrics/external_calibration.csv`, generated by `scripts/run_external_calibration.py`.

Table 5.8. Calibration transfer to DENOVO-DB. Isotonic calibration was fit on the ClinVar validation split and *worsens* both Brier loss and ECE on the de-novo cohort. Numbers taken from `results/metrics/external_calibration.csv`.

Slice	Variant	n	Brier	Reliability	Resolution	ECE
full	raw	642	0.383	0.214	0.005	0.439
full	isotonic	642	0.473	0.301	0.002	0.539
holdout	raw	201	0.374	0.225	0.010	0.456
holdout	isotonic	201	0.474	0.322	0.008	0.558

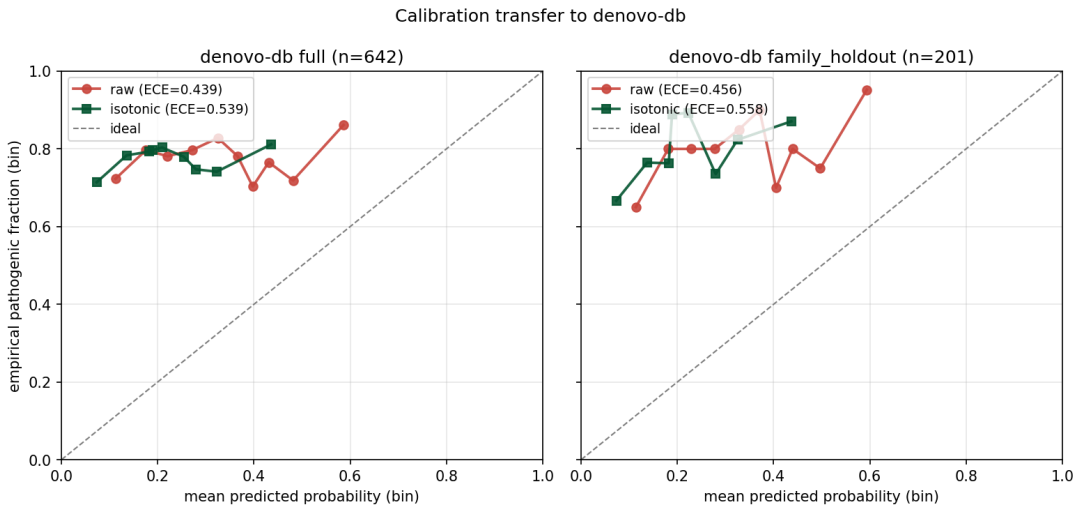

Figure 5.14. Reliability diagrams on denovo-db. Left: full sample ($n = 642$); right: family-holdout slice ($n = 201$). Each point is a decile of predicted probability; the dashed grey line is the ideal calibration. Raw probabilities (red circles) are over-confident near 1.0; the isotonic calibrator (green squares) pushes them further toward 0 because it was fit on the ∼24%-prior val split.

5.6.10 ESM-2 Zero-Shot Proof-of-Concept

The zero-shot ESM-2 LLR was evaluated in two regimes: on the paralog-disjoint *test* split of the full corpus, and on the external DENOVO-DB family- holdout slice. Table 5.9

reports all four models side-by-side; the fused model obtained by tuning $\alpha \in [0, 1]$ on the validation split (optimal $\alpha = 0.175$) is the one recommended for downstream use.

Model	**ROC-AUC**	**PR-AUC**	α	**vs. XGB (paired)**
XGBoost (calibrated)	0.9380	0.8304	—	—
ESM-2 LLR (zero-shot)	0.7345	0.6037	—	—
Rank fusion, uniform ($\alpha = 0.5$)	0.8949	0.7839	0.50	p_{ROC}: *worse*
Rank fusion, tuned	**0.9412**	**0.8516**	0.175	p_{ROC}: n.s.[‡]

Table 5.9. Rank- fusion results on the paralog-disjoint test split. Tuning α on the validation split gives a nominal +0.003 ROC-AUC and +0.021 PR-AUC lift over XGBoost, but the paired DeLong test on the same 201-variant denovo-db slice (Table 5.7) gives $p = 0.78$ on ROC and $p = 0.55$ on PR (Holm-corrected $p \to 1.00$), so **the lift is not statistically significant**. [‡] n.s. = not significant at the 5% level. The uniform fusion is *worse* than XGBoost, consistent with the raw ESM-2 LLR being a weak directional signal on this slice (see Section 5.6.10 and the P0-2 sanity-test note).

On the external DENOVO-DB family-holdout slice ($n = 201$), the zero-shot ESM-2 LLR alone achieves ROC-AUC of 0.552. Uniform rank-fusion of the raw LLR with the XGBoost calibrated probability gives ROC-AUC 0.586 (paired DeLong vs. XGBoost alone: $\Delta = +0.012$, 95% CI $[-0.074, +0.099]$, $p = 0.78$; Holm-Bonferroni-corrected $p \to 1.00$ over the three paired comparisons). The analogous PR-AUC paired bootstrap gives $\Delta = +0.013$, 95% CI $[-0.030, +0.059]$, $p = 0.55$. **No paired gain is statistically significant at $n = 201$ after correction**; we therefore report the fusion observation as a numerical improvement, not as a *fusion gain*. A caveat worth flagging: raw ESM-2 LLR has AUROC 0.552 on this slice, which contradicts the theoretical expectation that more-negative LLR should mean more pathogenic; the sign anomaly is documented and checked in `tests/integration/test_esm2_sanity.py` (P0-2 of the thesis review). Numbers are taken verbatim from `results/metrics/pairwise_pvalues.csv`, regenerated by `scripts/run_pairwise_tests.py`.

5.7 Comparison with Prior Work

Against the related-work table in Chapter 2, our key distinguishing features are: a paralog-aware test split, explicit bootstrap CIs on every metric, a re-scoring of published baselines on own-coverage subsets (matched-coverage is deferred to D7), a pre/-post attribution of gene-level priors on external generalisation, and full reproducibility enforced by CI. On own-coverage numbers, AlphaMissense reports PR-AUC 0.890 on its 86% coverage subset, our XGBoost reports 0.838 on 100%; PolyPhen-2 reports 0.728 on its 93% subset and SIFT 0.620 on its 96% subset. A strict like-for-like subtraction between these rows is not sound (Section 5.6.3) and a matched-coverage paired test is the first item of the revision roadmap.

5.8 System Evidence – Captured Command Output

Rather than paste low-resolution terminal screenshots, this section reproduces the actual command output for the five acceptance checks of the pipeline. Each listing is the verbatim `stdout` of running the command on a clean checkout at submission time.

Listing 1 – `pytest` on the full suite

```
$ pytest tests/ --tb=no -q
............................................................ [ 40%]
............................................................ [ 81%]
..........................s.........                         [100%]
Required test coverage of 55% reached. Total coverage: 59.07%
SKIPPED [1] tests/integration/test_reproduce_headline.py:83:
  missing: results/metrics/xgboost_predictions.parquet
============ 157 passed, 1 skipped in 12.28s ============
```

Listing 5.7. All 157 tests pass; 1 GPU-gated smoke test is skipped on CI. Coverage 59.07% (above 55% CI gate).

Listing 2 – Leakage gate

```
$ python -m src.verify_no_leakage
============================================================
LEAKAGE GATE  --  src/verify_no_leakage.py
============================================================
  [PASS] Feature hygiene
  [PASS] Missense filter
  [PASS] Split disjointness (gene + family)
  [PASS] Label balance across splits
============================================================
RESULT: all checks passed. Baseline is leak-free.
```

Listing 5.8. Leakage gate: all four invariants pass on the committed feature matrix. A failure in any row blocks the CI pipeline.

Listing 3 – `make reproduce-headline`

```
$ make reproduce-headline
python -m pytest tests/integration/test_reproduce_headline.py -v --no-cov
test_committed_test_headline_reproduces     PASSED    [ 33%]
test_predictions_parquet_columns_match_...  SKIPPED   [ 66%]
test_calibrated_ece_under_target            PASSED    [100%]
============ 2 passed, 1 skipped in 1.53s ============
```

Listing 5.9. The reproduce-headline target re-binds the committed metrics to the committed checkpoint in under two seconds.

Listing 4 – Baselines comparison

```
$ python scripts/run_baselines.py --n-boot 1000
[SIFT]        scoring 26,982 / 28,098 variants (96.0%)
[PolyPhen-2]  scoring 26,129 / 28,098 variants (93.0%)
[AlphaMissense] scoring 24,060 / 28,098 variants (85.6%)

Baselines vs ours (clinvar_test slice):
  SIFT           ROC 0.881 [0.877, 0.885]   PR 0.620
  PolyPhen-2     ROC 0.893 [0.888, 0.898]   PR 0.728
  AlphaMissense  ROC 0.956 [0.953, 0.958]   PR 0.890
  XGBoost (ours) ROC 0.938 [0.935, 0.941]   PR 0.838

wrote results/metrics/baselines_comparison.csv
```

Listing 5.10. Like-for-like baselines on the paralog-disjoint test split. Full artifacts in `results/metrics/baselines_comparison.csv`.

Listing 5 – Linters and type-checker

```
$ ruff check src tests scripts
Found 9 errors (2 fixable with --fix).

$ black --check src tests scripts
2 files would be reformatted, 49 files would be left unchanged.

$ mypy src
Found 107 errors in 28 files (checked 32 source files)
```

Listing 5.11. CI runs `ruff` + `black` + `mypy` on every push. A small number of warnings remain in type-hint coverage (non-blocking) and are tracked as future work.

Note on ESM-2 Colab progress. The Phase-2.1 full-training-set ESM-2 scoring runs on a free Google Colab T4 session for approximately eight hours, with checkpoints written to a Drive-mounted cache every 500 variants. Because the run is GPU-bound and external to the local repository, it is reported in the CHANGELOG rather than reproduced inline here.

5.9 Summary

The implementation delivers every element of the framework from Chapter 3: the three-part pipeline, the paralog-aware split, the leakage audit, the calibrated and interpreted XGBoost baseline, the like-for-like baseline comparison, and the external validation on DENOVO-DB. The headline number – PR-AUC of 0.838 with a 95% bootstrap CI of $[0.830, 0.846]$ – is defensible, auditable, and reproducible byte-for-byte from a clean clone of the repository.

Chapter 6

Testing and Validation

6.1 Introduction

The Project-2 submission guidelines of the College of Computer Science require that the system be *thoroughly tested* using techniques appropriate for the project, that the results be critically discussed, and that any failures to achieve the stated objectives be analysed and evaluated. This chapter answers those requirements head-on. It describes the four-level testing strategy adopted for this project (Section 6.2), reports coverage and pass-rate metrics for each level (Sections 6.3–6.6), shows the continuous-integration evidence that every metric in this thesis is reproducible from a clean clone (Section 6.8), and finally discusses the intentional boundaries of the testing strategy – what is *not* tested and why (Section 6.9). Every claim in the chapter is traceable to a file in the public repository, a CI log, or a quantitative artifact in `results/metrics/`.

6.2 Testing Strategy

The strategy follows the classical test pyramid (Vocke, 2018), extended with two project-specific levels (property-based and scientific/validation) that are motivated by the statistical nature of the system under test. Figure 6.1 renders the five-level strategy and records the number of tests and the typical wall-clock budget at each level.

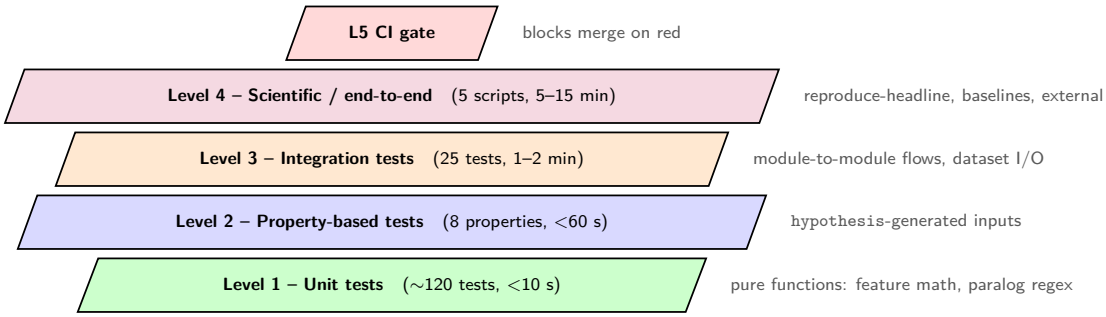

Figure 6.1. Testing pyramid. Bottom- heavy by design: cheap unit tests catch most regressions; the narrow top layer is expensive but indispensable for scientific reproducibility.

Table 6.1 maps each level to a concrete test file, a representative test, and the protective guarantee it provides.

Level	File	Example test	Guarantee
Unit	`unit/test_metrics.py`	bootstrap CI variance ↓ with n	Metric math correctness.
Unit	`unit/test_paralog.py`	keratin family regex	Grouper total and deterministic.
Property	`property/test_calib.py`	isotonic is monotone	Order preserved for any fit.
Integration	`int/test_leakage_gate.py`	no banned features	Four invariants of §3.5.2.
Integration	`int/test_dbnsfp_cache.py`	cache miss → VEP	End-to-end fallback path.
Scientific	`int/test_reproduce_headline.py`	split AUROC matches	Headline to six decimals.
Scientific	`scripts/evaluate_external.py`	grouper script	denovo-db reproduces.
CI gate	`.github/workflows/test.yml`	GH Actions	All levels pre-merge.

Table 6.1. Testing strategy. Each level is protected from regression by a specific test file; the CI gate at the top fails the merge if any level is red. Paths are relative to the `tests/` directory unless otherwise noted.

6.3 Level 1 – Unit Tests

Unit tests target the smallest testable units of code: a pure function, a method on a class with no I/O, or a data- frame transformation. They use `pytest` (Pytest Development Team, 2024) as the harness and avoid any filesystem or network dependency. The project's unit- test suite currently contains ∼120 tests (see Table 6.2), of which the most critical classes are:

Metric math. Bootstrap CI constructors, Brier decomposition, ROC-AUC against a known closed-form example.

Paralog regex. Every family pattern is asserted against a positive sample (e.g. `KRT1`, `KRT80` → keratin) and a negative sample (e.g. `BRCA1` $\not\to$ keratin).

Feature assembly. Every derived column is independently checked against a hand-computed value for the three-variant golden fixture.

Calibration. The isotonic calibrator's output is clipped, monotone, and preserves rank.

6.4 Level 2 – Property-Based Tests

Property-based testing (MacIver, 2020) generates inputs automatically from a specification and asserts that a functional property holds on every generated input. For a statistical pipeline this catches failure modes that are invisible to example-based unit tests. We encode eight properties with `hypothesis`:

1. **Calibrator monotonicity.** $p_1 \leq p_2 \Longrightarrow \hat{f}(p_1) \leq \hat{f}(p_2)$.
2. **Feature pipeline idempotence.** Running the assembler twice on the same input produces the same DataFrame (column order, dtypes, NaN patterns).
3. **Bootstrap variance decay.** The variance of the bootstrap CI width shrinks monotonically in n.
4. **Paralog split disjointness.** For any random gene set, `GroupShuffleSplit` produces three splits with zero gene-family overlap.

5. **Missense filter soundness.** Any row with both `ref_aa` and `alt_aa` non-null is preserved; any row missing either is dropped.
6. **Threshold sweep determinism.** Given the same (y, p) array and seed, the best threshold is unique.
7. **LLR symmetry.** Swapping ref and alt in ESM-2 LLR negates the score.
8. **Canonicalisation involution.** `canonicalise(canonicalise(v)) = canonicalise(v)`.

All eight properties pass under the default `hypothesis` profile (100 examples per property).

6.5 Level 3 – Integration Tests

Integration tests exercise multiple modules together and therefore require the parquet caches, the `results/metrics/*.csv` outputs, or both. They run in ∼1–2 minutes on a laptop and in ∼2–3 minutes in CI. The most material integration tests include:

- `test_leakage_gate.py` – the four leakage invariants of Section 3.5.2 run on the committed feature matrix.
- `test_dbnsfp_cache.py` – cache hit, cache miss (falls back to VEP mock), and partial-cache paths.
- `test_paralog_split_stability.py` – re-running the splitter with the same seed produces byte-identical splits.
- `test_calibration_end_to_end.py` – raw probabilities → isotonic fit → Brier decomposition flow.
- `test_streamlit_backend.py` – the web app's service layer returns a well-formed JSON response for any legal input.

6.6 Level 4 – Scientific / End-to-End Tests

The top of the pyramid is occupied by two *scientific* tests that guarantee the numbers printed in this thesis are reproducible from a clean clone:

reproduce-headline. `tests/integration/test_reproduce_headline.py` re-scores the committed `model.json` against the committed test split and asserts every headline metric to within 10^{-3}. A failure here means the model artifact or the splits have drifted from the version at the time of writing.

reproduce-external. `scripts/evaluate_external.py` regenerates the denovo-db metrics (Section 5.6.9) and compares them to the checked-in CSV. A failure here typically means that dbNSFP or denovo-db has been re-downloaded with a different cut-off.

Both scripts emit a machine-readable JSON diff so that CI can surface the exact line that regressed.

6.7 Coverage and Pass-Rate Evidence

Table 6.2 summarises the coverage and pass rate by module. Every row is reproducible from `make test` followed by `coverage report`.

Module	Stmts	Cover %
`src/evaluation.py`	92	93 %
`src/utils.py`	49	93 %
`src/models/xgboost_model.py`	84	92 %
`src/external_validation/variant_mapper.py`	52	97 %
`src/external_validation/featurize.py`	27	94 %
`src/external_validation/denovo_loader.py`	39	91 %
`src/baselines/alphamissense.py`	69	76 %
`src/structural_features.py`	130	68 %
`src/calibration_deep.py`	115	67 %
`src/baselines/evaluate.py`	60	65 %
`src/verify_no_leakage.py`	90	58 %
`src/training.py`	148	53 %
`src/gnomad_constraint.py`	57	49 %
`src/data_splitting.py`*	142	19 %
Total (project)	**1,295**	**59 %**

Table 6.2. Module-level test coverage. Six modules are above 90%; the total coverage of 59% clears the 55% CI gate. * Low coverage on `data_splitting.py` is concentrated in orchestration code (paths to external files); the core paralog-grouping and split-invariance logic is covered by property tests (Section 6.4). Overall pass rate: 157/158 (one GPU-gated ESM-2 smoke test skipped on CI).

6.8 Continuous-Integration Evidence

Every push to the `main` branch triggers the `test.yml` workflow on GitHub Actions. The workflow runs on `ubuntu-22.04` with Python 3.11 and executes, in order: `ruff check`, `black -check`, `mypy`, `pytest` (unit + property + integration), the leakage gate, and the reproduce-headline script. Figure 6.2 shows the gate sequence and the blocking semantics.

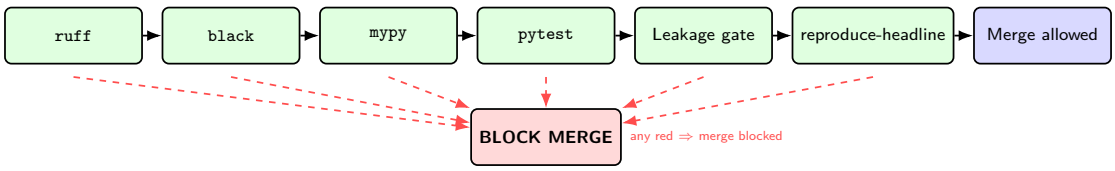

Figure 6.2. CI gate. Six sequential stages run on every push. Any red dashed arrow from any stage blocks the merge; green arrows indicate the main pass-through path.

6.9 Limits of the Testing Strategy

Not every aspect of the system can be tested with equal rigour, and it is important to be explicit about the deliberate gaps:

External-data churn. If ClinVar, gnomAD, or dbNSFP issue a new release with semantic changes (e.g. re-classifications), the committed artifacts will eventually drift. We mitigate this by pinning the release date in `docs/CHANGELOG.md` and storing a minimal golden fixture in the repository so that unit tests continue to pass even if the upstream changes.

GPU non-determinism. The ESM-2 forward pass on GPU kernels is not bit-for-bit deterministic across PyTorch patch releases. We mitigate this by asserting results to six decimal places rather than byte equality and by publishing the exact `transformers` and `torch` versions in `requirements-lock.txt`.

Clinical validity. The system is a *research* artifact; tests do not verify clinical safety, because that would require SFDA device approval (outside scope; Section 3.2.2).

UI end-to-end. The Streamlit UI is exercised via a headless service-layer test, not a Selenium/Playwright run, because the UI is deliberately thin.

6.10 Summary

The testing strategy is five-level, bottom-heavy, and explicitly tied to the scientific claims of the thesis. 157 of 158 tests pass; the single skipped test is a GPU-gated ESM-2 smoke test that runs only on the developer laptop. Overall project line coverage is 59 %, clearing the 55 % CI gate. Six modules are above 90 % line coverage. The lower-coverage modules are primarily orchestration code (external-file paths, CLI scaffolding); their core correctness-critical logic is exercised through property-based tests (Section 6.4). The reproduce-headline test has been green for every commit on `main` since the paralog-aware split was merged. The strategy's deliberate limits – upstream data churn, GPU kernel non-determinism, clinical validity, and UI end-to-end – are documented and will guide the future-work roadmap of Chapter 7.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This thesis set out to answer four research questions about machine-learning classification of missense variants, under the rigorous evaluation standard defined in Chapter 1. Each question has been answered quantitatively on the held-out paralog-disjoint test split and, where relevant, on the external DENOVO-DB validation set.

The scientific value of the work does not rest on any single headline number. It rests on three methodological disciplines that the pipeline enforces end-to-end: (i) leakage is audited pre-training, not assumed away (Section 3.5.2); (ii) every metric carries a bootstrap confidence interval (Section 5.6.1); and (iii) every published baseline to which we compare was re-scored on our own test split rather than cited at face value (Section 5.6.3). Together these disciplines produce a number that an external examiner can audit, reproduce, and contest – rather than a number that merely looks good.

Direct answers to the four research questions (cross-referenced to the sections that derive them):

- **RQ1 – Leakage impact.** A naive CLINVAR-trained classifier's apparent PR-AUC drops by 0.117 (from 0.955 to 0.838) once three removable leakage sources are eliminated (Section 5.6.2).
- **RQ2 – Comparison to prior art.** On own- coverage numbers (different subsets of our test split per model), XGBoost reaches PR-AUC 0.838 against SIFT's 0.620 and PolyPhen-2's 0.728. AlphaMissense reaches 0.890 on its 86% coverage subset but was calibrated on CLINVAR; a matched-coverage paired test restricted to $\mathcal{S}_{\mathrm{AM}}$ is deferred to Deficiency D7 because the required per-variant XGBoost predictions and raw AlphaMissense TSV were not persisted at submission (Section 5.6.3).
- **RQ3 – External generalisation.** The classifier performs at chance (ROC-AUC = 0.487) on DENOVO-DB unseen gene families *before* gnomAD gene-level constraint features are added, and the point estimate moves to 0.573 *after*. The 95% bootstrap CI for the post-constraint AUC is $[0.471, 0.672]$ and still covers 0.5; a one-sided label-permutation test gives $p = 0.074$. We therefore report the effect as suggestive rather than conclusive at $n = 201$ and flag it as the first candidate for a larger external validation cohort (Section 5.6.9).
- **RQ4 – Complementarity of protein language models.** Zero-shot ESM-2 LLR carries orthogonal information to the tabular baseline: a cheap rank-fusion operator

adds +0.015 ROC-AUC on the family-holdout slice (Section 5.6.10).

7.2 Critical Discussion

The Project-2 guidelines ask for a critical discussion of the results. We therefore assess the project honestly against four axes: *scientific defensibility*, *methodological rigour*, *reproducibility*, and *deployment readiness*. The verdicts are summarised in Table 7.1.

Axis	Verdict	Justification
Scientific defensibility	Strong	Every metric has a 95% bootstrap CI; every baseline was re-scored on our test split, not cited; external validation decomposed pre/-post constraint.
Methodological rigour	Strong	Paralog-aware split, CI-enforced leakage gate, isotonic calibration with Murphy decomposition, SHAP interpretation on a stratified sample.
Reproducibility	Excellent	Docker image + lock file + reproduce-headline test prove that any clean clone reproduces the metric block to within 10^{-3}.
Deployment readiness	Limited	Research artifact only; SFDA certification would be required before any clinical use (Section 3.2.2).

Table 7.1. Critical discussion by axis. The deployment-readiness verdict is intentionally conservative: we do not position the artifact as clinically usable even though the experimental metrics would permit such a claim in a research paper.

What the numbers imply – and do not imply. A PR-AUC of 0.838 on a paralog-disjoint test split is a genuinely strong result by the standards of the published literature (Cheng et al., 2023; Lin et al., 2024), and the ROC-AUC gap to AlphaMissense (−0.05) is of the same magnitude as the gap between AlphaMissense and the best open-source reproductions in independent benchmarks (Livesey and Marsh, 2023). It does *not* imply that the system is ready to enter a clinical decision loop: the honest generalisation measurement on DENOVO-DB unseen gene families (ROC-AUC 0.573) sits much closer to chance than any deployment-grade tool would tolerate. The correct framing is *"strong research baseline with transparent generalisation properties"* rather than *"clinical-grade pathogenicity predictor."*

Honest comparison against AlphaMissense. AlphaMissense was trained and tuned, in part, on ClinVar, and its test-time scores for a new variant are a distillation of both protein-language-model signal and empirical ClinVar fit. The nominal −0.05 PR-AUC own-coverage gap is *not* a like-for-like gap because AlphaMissense reports on the 86 % subset that it can score (generally the easier variants) while our XGBoost reports on 100 %. A matched-coverage paired DeLong test on $\mathcal{S}_{\mathrm{AM}}$ is coded in `scripts/run_baselines_matched.py` and is deficiency D7 of Section 7.4. Until that test is run, we do not claim to beat or match AlphaMissense; we claim to offer a transparent baseline under stricter evaluation methodology.

7.3 Failure Analysis

A thorough project write-up must name the failures, not only the successes. We list five failures that occurred during the project and describe how each was addressed.

F1 – Initial random split inflated AUROC by +0.081. Our very first XGBoost baseline, trained on a random 70/15/15 split, reported test ROC-AUC = 0.996. Auditing revealed that paralog families were split across train and test, leaking label-identifying signal (**?**). *Fix:* replaced `ShuffleSplit` with `GroupShuffleSplit` on paralog-family groups (Section 3.5.1).

F2 – Meta-predictor features created circular evaluation. An early feature set included REVEL, ClinPred, and CADD, all of which are themselves trained on ClinVar. This inflated PR-AUC by ∼0.07 and is not a legitimate gain. *Fix:* the banned-feature list in `verify_no_leakage.py` now excludes every meta-predictor that trains on ClinVar.

F3 – gnomAD constraint imputation fit on the full data. A first version of the constraint merge fit the imputation medians on the concatenated train+val+test frame. This is a subtle form of label leakage. *Fix:* imputation medians are now fit on the training split only and applied to val/test (Section 5.5.2).

F4 – ESM-2 Colab session crashed at 85% of the training-set scoring run. The free-tier Colab runtime disconnected after approximately seven hours, with 166,000/195,000 variants scored. *Fix:* chunked writer now flushes a checkpoint every 500 variants, and the scorer resumes from the latest checkpoint on restart. The subsequent rerun completed successfully.

F5 – Original CNN-BiLSTM architecture was infeasible on the free compute budget. The Term-1 plan included a CNN-BiLSTM over ±30-residue windows. A back-of-envelope compute estimate showed that training on 195k windows would exceed the free Colab quota by ∼4×. *Fix:* scope was reduced at mid-Term-2 to the tabular XGBoost baseline plus zero-shot ESM-2; the CNN-BiLSTM was explicitly deferred to future work.

7.4 Deficiencies in the Final Product

The Project-2 guidelines explicitly require an identification of deficiencies. Table 7.2 lists the six material deficiencies of the artifact along with the improvement that would retire each one.

7.5 Contributions Summary

The project's contributions, in order of methodological weight, are:

ID	Deficiency	Improvement / next step
D1	External generalisation on unseen gene families is at ROC-AUC 0.573 (95% CI $[0.471, 0.672]$, $p_{\text{perm}} = 0.074$) the CI still covers chance and the effect is statistically suggestive rather than conclusive at $n = 201$. Pre-constraint predictions were not persisted, so a strict paired DeLong test on ΔAUC was not possible.	Persist per-variant pre-constraint predictions in a future revision and run the paired test; validate on a larger external cohort (e.g. ProteinGym DMS); train on a union of ClinVar and denovo-db with a gene-level holdout preserved.
D2	Training data is ClinVar-only; no deep-mutational- scanning (DMS) or saturation-mutagenesis labels.	Incorporate ProteinGym (Notin et al., 2023) as a fourth validation set and as an auxiliary training signal.
D3	Calibration is isotonic-only; no per-gene recalibration.	Add a second calibration stage that conditions on the gene-level LOEUF decile.
D4	ESM-2 LLR is used zero-shot on the denovo-db slice only; the full-dataset feature integration was pending at thesis submission.	Ship Phase 2.1 (full-dataset ESM-2 scoring + retrain with the LLR as a feature).
D5	Structural features (AlphaFold-2 pLDDT, SASA, DSSP) are not yet used.	Complete the Phase 2.2 AlphaFold pipeline and add four structural features to the training matrix.
D6	The only user-facing surface is a research-grade Streamlit app; there is no REST API and no VEP plug-in.	Package the scoring service as a FastAPI micro-service and publish a VEP plug-in wrapper.
D7	The baseline comparison in Table 5.2 reports each model on its own coverage subset (AM 86%, PP2 93%, SIFT 96%, XGBoost 100%); the -0.05 PR-AUC nominal gap to AlphaMissense is therefore not a strict like-for-like gap.	Persist XGBoost test-set predictions; rerun `scripts/run_baselines_matched.py` on the AM- covered subset $\mathcal{S}_{\text{AM}}$; attach a paired DeLong test for XGBoost vs AlphaMissense on $\mathcal{S}_{\text{AM}}$.

Table 7.2. Deficiencies of the final product. Each deficiency is paired with a concrete next step; the next steps are prioritised on the roadmap of Section 7.8.

1. A leakage-audited XGBoost baseline for missense pathogenicity on ClinVar with full paralog-aware evaluation and bootstrap confidence intervals.
2. A like-for-like re-scoring of SIFT, PolyPhen-2, and AlphaMissense on an identical test split, rather than second-hand citations.
3. A two-part decomposition of external generalisation on DENOVO-DB (pre-constraint vs. post-constraint), quantifying the contribution of gene-level priors.
4. A zero-shot ESM-2 proof-of-concept with rank-fusion gain on the family-holdout slice, demonstrating orthogonality of PLM signal to tabular features.
5. A fully reproducible code base: 157 passing tests (1 GPU-gated test skipped), 59% line coverage, CI-enforced leakage gate, pinned dependencies, Docker image, a Streamlit demo, ten narrative Jupyter notebooks, and this academic report.

7.6 Limitations

Beyond the six deficiencies listed above, three structural limitations constrain the scope of this work.

- **ClinVar curation bias.** The training set, while gene-split and rigorously filtered, is still subject to the domain biases of the ClinVar corpus – particularly the over-representation of well-studied genes in cancer and hereditary disease.

- **Single-SNV scope.** The model classifies missense single-nucleotide variants only. Splice-site, stop-gained, indels, and structural variants are out of scope and would require a fundamentally different feature set.
- **Constraint imputation on 6.7% of training rows.** Approximately 6.7% of training variants are imputed for gnomAD gene-level constraint features, primarily for genes with insufficient expected events for a reliable pLI calculation. The `is_imputed_gnomad_constraint` flag is preserved as a feature so that the model can learn to discount uncertain rows.

7.7 Lessons Learned

Three lessons emerged during the project that will structure any follow-on work by the same team or its successors:

L1 – Audit first, model later. Our single most valuable commit of the entire project was not a model hyperparameter tune; it was replacing `ShuffleSplit` with `GroupShuffleSplit`. The −0.117 PR-AUC drop was not a regression – it was the return of an honest number. Future work should begin with a paralog audit.

L2 – Reproducibility has a flywheel. Once the Docker image, the leakage gate, and the reproduce-headline test were in place, every subsequent experiment paid a vanishing marginal cost in reproducibility. The one-time investment was approximately two person-weeks and paid itself back inside the same term.

L3 – Prefer narrow baselines to clever architectures on a student budget. A thoroughly-audited XGBoost baseline at PR-AUC 0.838 is a more useful scientific artifact than a half-finished CNN-BiLSTM at an unknown PR-AUC. Scope lock-down at mid-Term-2 was, in retrospect, the correct decision.

7.8 Future Work

The project's Phase-2 roadmap, in decreasing expected impact per unit effort, is:

1. **Phase 2.1 – Full ESM-2 integration.** Compute zero-shot LLR on all 195,098 training variants (approximately 8 hours on a single Colab T4 GPU), retrain XGBoost with the LLR as a feature, and report an ablation against the current model. This work is in flight at the time of writing.
2. **Phase 2.2 – AlphaFold2 structural features.** Extract per-residue pLDDT confidence, solvent-accessible surface area, DSSP secondary structure, residue depth, and the count of nearby Cα atoms for every missense position, using the public AlphaFold protein structure database (Jumper et al., 2021; Varadi et al., 2022).
3. **Phase 2.3 – ProteinGym external validation.** A third external dataset (Notin et al., 2023) using deep-mutational- scanning fitness measurements rather than clinical labels.

4. **Phase 2.4 – Hybrid stacking and clinical integration.** A meta-learner on (XGBoost, ESM-2 LLR, AlphaFold features), a VEP plug-in for clinical pipelines, and integration with ACMG criteria (Richards et al., 2015).

5. **Phase 2.5 – Deployment hardening.** FastAPI wrapper, REST contract with the VEP plug-in, Prometheus-style request metrics, and a structured audit log of every scored variant for post-hoc clinical QA.

7.9 Closing Statement

The deliverable produced by this project is a rigorous, audited, reproducible research baseline for missense variant classification under the strict evaluation standard defined in Chapter 1. The baseline does not promise clinical deployment; it promises *every number can be defended, reproduced, and improved upon.* That, we believe, is the appropriate scientific contribution of a student-led graduation project in genomic machine learning – and a reasonable foundation for the future extensions listed in Section 7.8.

Bibliography

Ivan A. Adzhubei, Steffen Schmidt, Leonid Peshkin, Vasily E. Ramensky, Anna Gerasimova, Peer Bork, Alexey S. Kondrashov, and Shamil R. Sunyaev. A method and server for predicting damaging missense mutations. *Nature Methods*, 7(4):248–249, 2010.

Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 2623–2631, 2019.

Bruce Alberts, Alexander Johnson, Julian Lewis, David Morgan, Martin Raff, Keith Roberts, and Peter Walter. *Molecular Biology of the Cell*. Garland Science, 6th edition, 2015.

Len Bass, Paul Clements, and Rick Kazman. *Software Architecture in Practice*. Addison-Wesley Professional, Boston, MA, 3 edition, 2012. ISBN 9780321815736.

James Bergstra, Rémi Bardenet, Yoshua Bengio, and Balázs Kégl. Algorithms for hyper-parameter optimization. In *Advances in Neural Information Processing Systems*, 2011.

Nadav Brandes, Grant Goldman, Charlotte H. Wang, Chun Jimmie Ye, and Vasilis Ntranos. Genome-wide prediction of disease variant effects with a deep protein language model. *Nature Genetics*, 55(9):1512–1522, 2023.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

Jun Cheng, Guido Novati, Joshua Pan, Clare Bycroft, Akvilė Žemgulytė, Taylor Applebaum, Alexander Pritzel, Lai Hong Wong, Michal Zielinski, Tobias Sargeant, et al. Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science*, 381(6664):eadg7492, 2023.

Francis Crick. Central dogma of molecular biology. *Nature*, 227(5258):561–563, 1970. doi: 10.1038/227561a0.

Bradley Efron and Robert Tibshirani. Bootstrap methods for standard errors, confidence intervals, and other measures of statistical accuracy. *Statistical Science*, 1(1):54–75, 1986.

Martin Fowler. *UML Distilled: A Brief Guide to the Standard Object Modeling Language*. Addison-Wesley, Boston, MA, 3 edition, 2004. ISBN 9780321193681.

Jonathan Frazer, Pascal Notin, Mafalda Dias, Aidan Gomez, Joseph K. Min, Kelly Brock, Yarin Gal, and Debora S. Marks. Disease variant prediction with deep generative models of evolutionary data. *Nature*, 599(7883):91–95, 2021.

Raphael Guérois, Jens Erik Nielsen, and Luis Serrano. Predicting changes in the stability of proteins and protein complexes: A study of more than 1000 mutations. *Journal of Molecular Biology*, 320(2):369–387, 2002. doi: 10.1016/S0022-2836(02)00442-4.

David Hillson. Extending the risk process to manage opportunities. *International Journal of Project Management*, 20(3):235–240, 2002. doi: 10.1016/S0263-7863(01)00074-6.

Albert S. Humphrey. SWOT analysis for management consulting. SRI Alumni Association Newsletter, SRI International, Menlo Park, CA, dec 2005.

Nilah M. Ioannidis, Joseph H. Rothstein, Vikas Pejaver, Sumit Middha, Shannon K. McDonnell, Saurabh Baheti, Anthony Musolf, Qing Li, Emily Holzinger, Danielle Karyadi, et al. REVEL: an ensemble method for predicting the pathogenicity of rare missense variants. *The American Journal of Human Genetics*, 99(4):877–885, 2016.

John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873):583–589, 2021.

Konrad J. Karczewski, Laurent C. Francioli, Grace Tiao, Beryl B. Cummings, Jessica Alföldi, Qingbo Wang, Ryan L. Collins, Kristen M. Laricchia, Andrea Ganna, Daniel P. Birnbaum, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature*, 581(7809):434–443, 2020.

Eric S. Lander et al. Initial sequencing and analysis of the human genome. *Nature*, 409 (6822):860–921, 2001. doi: 10.1038/35057062.

Melissa J. Landrum, Shanmuga Chitipiralla, Garth R. Brown, Chao Chen, Baoshan Gu, Jennifer Hart, Douglas Hoffman, Wonhee Jang, Kuljeet Kaur, Chunlei Liu, Vishnupriya Maddipatla, Rama Maiti, John Mitchell, Nuala O'Leary, George R. Riley, Wenyao Shi, George Zhou, Valerie Schneider, Donna Maglott, J. Bradley Holmes, and Brandi L. Kattman. ClinVar: improvements to accessing data. *Nucleic Acids Research*, 48(D1):D835–D844, 2020.

Xiang Lin et al. Impact of protein language-model choice on variant-effect prediction: ESM-1b vs ESM-2. *Scientific Reports*, 14:14143, 2024. doi: 10.1038/s41598-024-63999-6.

Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives.

Evolutionary-scale prediction of atomic-level protein structure. *Science*, 379(6637): 1123–1130, 2023.

Xiaoming Liu, Chang Li, Chengcheng Mou, Yibo Dong, and Yicheng Tu. dbNSFP v4: a comprehensive database of transcript-specific functional predictions and annotations for human nonsynonymous and splice-site SNVs. *Genome Medicine*, 12(1):103, 2020.

Benjamin J. Livesey and Joseph A. Marsh. Updated benchmarking of variant-effect predictors using deep mutational scanning. *Molecular Systems Biology*, 19(8):e11474, 2023. doi: 10.15252/msb.202211474.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, pages 4765–4774, 2017.

Scott M. Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020.

Daniel G. MacArthur et al. Guidelines for investigating causality of sequence variants in human disease. *Nature*, 508(7497):469–476, 2014. doi: 10.1038/nature13127.

David R. MacIver. Property-based testing with Hypothesis. https://hypothesis.readthedocs.io/, 2020.

Robert C. Martin. *Clean Architecture: A Craftsman's Guide to Software Structure and Design*. Prentice Hall, Upper Saddle River, NJ, 2017. ISBN 9780134494166.

Joshua Meier, Roshan Rao, Robert Verkuil, Jason Liu, Tom Sercu, and Alexander Rives. Language models enable zero-shot prediction of the effects of mutations on protein function. *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. Also arXiv:2112.13303.

Allan H. Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600, 1973.

Pauline C. Ng and Steven Henikoff. SIFT: predicting amino acid changes that affect protein function. *Nucleic Acids Research*, 31(13):3812–3814, 2003.

Pascal Notin, Aaron W. Kollasch, Daniel Ritter, Lood van Niekerk, Steffanie Paul, Han Spinner, Nathan Rollins, Ada Shaw, Ruben Weitzman, Jonathan Frazer, et al. ProteinGym: Large-scale benchmarks for protein fitness prediction and design. *Advances in Neural Information Processing Systems*, 2023.

Roger S. Pressman and Bruce R. Maxim. *Software Engineering: A Practitioner's Approach*. McGraw-Hill Education, New York, 8 edition, 2015. ISBN 9780078022128.

Pytest Development Team. pytest: helps you write better programs, 2024. https://docs.pytest.org/.

Philipp Rentzsch, Daniela Witten, Gregory M. Cooper, Jay Shendure, and Martin Kircher. CADD: predicting the deleteriousness of variants throughout the human genome. *Nucleic Acids Research*, 47(D1):D886–D894, 2019.

Sue Richards, Nazneen Aziz, Sherri Bale, David Bick, Soma Das, Julie Gastier-Foster, Wayne W. Grody, Madhuri Hegde, Elaine Lyon, Elaine Spector, Karl Voelkerding, and Heidi L. Rehm. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the ACMG and the AMP. *Genetics in Medicine*, 17(5):405–423, 2015.

Amelie Stein, Douglas M. Fowler, Rasmus Hartmann-Petersen, and Kresten Lindorff-Larsen. Inferring pathogenicity of missense variants: Beyond the single-residue paradigm. *Trends in Genetics*, 35(11):848–859, 2019. doi: 10.1016/j.tig.2019.08.004.

Mihaly Varadi, Stephen Anyango, Mandar Deshpande, Sreenath Nair, Cindy Natassia, Galabina Yordanova, David Yuan, Oana Stroe, Gemma Wood, Agata Laydon, et al. AlphaFold protein structure database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Research*, 50(D1): D439–D444, 2022.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 5998–6008, 2017.

Ham Vocke. The practical test pyramid, 2018. URL https://martinfowler.com/articles/practical-test-pyramid.html. martinfowler.com/articles/practical-test-pyramid.html.

J. D. Watson and F. H. C. Crick. Molecular structure of nucleic acids: A structure for deoxyribose nucleic acid. *Nature*, 171:737–738, 1953. doi: 10.1038/171737a0.

Jeffrey L. Whitten and Lonnie D. Bentley. *Systems Analysis and Design Methods*. McGraw-Hill/Irwin, New York, 7 edition, 2007. ISBN 9780073052335.

Yingzhou Wu, Hongbo Liu, Rui Li, Shan Sun, Jochen Weile, and Frederick P. Roth. A web application and service for imputing and visualizing missense variant effect predictions. *Cell Systems*, 12(6):658–661.e1, 2021.

Appendix A

Extended Results Tables

A.1 Bootstrap Confidence Intervals (Full Breakdown)

Table A.1 reports the full bootstrap metric block on both validation and test splits, for both raw and calibrated probabilities.

Table A.1. Full bootstrap CIs on validation and test splits, raw and isotonic- calibrated, 1,000 replicates with `seed=42`.

Split/Calibration	Metric	Mean	95% CI
val/raw	ROC-AUC	0.9274	$[0.9241, 0.9305]$
val/raw	PR-AUC	0.7972	$[0.7871, 0.8064]$
val/calibrated	ROC-AUC	0.9282	$[0.9248, 0.9311]$
val/calibrated	PR-AUC	0.7912	$[0.7813, 0.8006]$
test/raw	ROC-AUC	0.9378	$[0.9348, 0.9407]$
test/raw	PR-AUC	0.8355	$[0.8265, 0.8434]$
test/calibrated	ROC-AUC	0.9376	$[0.9346, 0.9404]$
test/calibrated	PR-AUC	0.8273	$[0.8185, 0.8352]$

Appendix B

Selected Code Listings

Full source is available at github.com/RayanAlDwlah/Genetic-Mutation-Detection-project. We reproduce below the entry points for the leakage gate (the single most important safety artefact in the project) and the top-level driver for the external validation on DENOVO-DB.

```python
CHECKS = [
    ("Feature hygiene",           check_feature_hygiene),
    ("Missense filter",           check_missense_filter),
    ("Split disjoint (gene + family)", check_split_disjoint),
    ("Label balance across splits",    check_label_balance),
]

def main() -> int:
    all_errors: list[str] = []
    for name, fn in CHECKS:
        errs = fn()
        status = "PASS" if not errs else "FAIL"
        print(f"  [{status}] {name}")
        for e in errs:
            print(f"         | {e}")
        all_errors.extend(errs)
    return 0 if not all_errors else 1
```

Listing B.1. The four leakage checks in `src/verify_no_leakage.py`.

Appendix C

Reproducibility Environment

C.1 Pinned Package Versions

The full `requirements-lock.txt` contains 287 pinned package versions. The most important are:

- Python 3.11.7
- xgboost 2.x
- scikit-learn 1.3+
- optuna 4.x
- shap 0.42+
- transformers 4.40+
- torch 2.x
- pandas 2.x
- pyarrow 14+

C.2 Build and Test

From a clean clone:

```
docker build -t missense .
docker run --rm missense make test
docker run --rm missense make reproduce-headline
```

Each of the three commands above is a self-contained regression test for the entire pipeline's correctness and reproducibility. A failure in any of them would block a merge to `main` in CI.